A guide to Conformal Prediction

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April 22, 2026

Remark:

This report was completed under the supervision of Professor Tarek Ould-Bachir for the course INF8900, "Lectures dirigées génie info. génie logiciel" at Polytechnique Montréal. I am deeply grateful to Professor Ould-Bachir for providing me with the opportunity to explore the fascinating subject of Conformal Prediction. I would also like to extend my sincere thanks to Professor Aymeric Dieuleveut of École Polytechnique (l'X), who introduced me to Conformal Prediction during the Hi!Paris Summer School in 2025.

Note that some structuring documents for this report are the work of Anastasios Angelopoulos (1), the slides from Aymeric Dieuleveut (2), the course by Ryan Tibshirani (3), and the great article by Jing Lei (4). Each time, I tried to improve the work by providing intuitive examples that are accessible on the GitHub page, making links to other works and focusing on what is essential to understand the underlying concept without neglecting the theoretical rigor. Finally, going further by describing novel and innovative approaches.

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1 Introduction

Conformal Prediction has gained significant attention in recent years. It is viewed as a statistical approach for improving the robustness of predictive models. For a long time, predictive modeling was dominated by statisticians and operated in settings where only a small, carefully selected number of data points was available. These data points formed datasets that were required to satisfy several assumptions in order to apply techniques such as regression models. Some of these assumptions include: (A) the data points must be independent and identically distributed (*i.i.d.*), (B) they should follow a specific distribution (e.g., a normal distribution), and (C) they must satisfy the homoscedasticity constraint (i.e., constant variance). Following breakthroughs in computational hardware, such as GPUs, traditional predictive models were increasingly overshadowed by Deep Learning methods (i.e., neural networks). These techniques require large datasets, and their complexity and non-linearity allow them to achieve strong predictive performance without relying on the aforementioned assumptions, which are often unrealistic in such settings. As a result, the machine learning community shifted its focus away from theoretical guarantees toward empirical performance. However, this shift came with significant drawbacks. In particular, it led to a loss of interpretability (due to black-box models) and a lack of guarantees regarding the confidence of predictions.

Therefore, to recover this capability in a setting with minimal assumptions on the data, a distribution-free, model-agnostic approach with strong theoretical foundations is needed. Conformal Prediction provides such a solution by reintroducing statistical guarantees into modern machine learning.

In this report, we begin by introducing the theoretical foundations of Conformal Prediction (Section 2). We then present one of the most widely used approaches, Split Conformal Prediction (Section 3). Next, we describe another important method, Full Conformal Prediction (Section 4), followed by a highlight of key techniques and challenges in Conformal Prediction (Section 5). We then explore more advanced methods and approaches (Section 6), including niche techniques (Section 6.5, and 6.6). Finally, we examine the potential of Conformal Prediction through three case studies (Section 7). But, first we start with a small History of Conformal Prediction.

1.1 History of Conformal Prediction

Conformal prediction emerged in the late 1990s through the work of Vladimir Vovk (a student of Andrey Kolmogorov), Alexander Gammerman, Craig Saunders and Vladimir Vapnik (1). They introduced a framework for distribution-free uncertainty quantification using p-values to construct prediction sets with guaranteed error rates. Early theoretical breakthroughs in the early 2000s established key properties, while split conformal prediction (with Harris Papadopoulos) made the method computationally efficient. Around the same time, Glenn Shafer coined the term "conformal predictor". A major turning point came with the popularization of conformal prediction in the United States by Jing Lei and Larry Wasserman, whose worked brought the framework to a wider audience and inspired further

advances by researchers such as Emmanuel Candès, Ryan Tibshirani and others. Recent research has broadened applications, some of them are described in this report.

2 Foundation

2.1 Defining the goal

The main conformal prediction idea is to try to create guarantees with respect to the range of values outputed by a predictive model, given a certain dataset. We have a collection of random values $(X_i, Y_i) \sim P$ for $i = 1, \ldots, n$, with $(X_i, Y_i) \in \mathcal{X} \times \mathcal{Y}$. In general, we have $\mathcal{X} \in \mathbb{R}^n$ and $\mathcal{Y} \in \mathbb{R}$. Then, we select an error level $\alpha \in (0, 1)$ and want our prediction on a new point to fall within a confidence region with probability of at least $1 - \alpha$.

Main Goal

Create a prediction band $\hat{C}_n$, that will be able to predict Y_{n+1} from X_{n+1} with a certain confidence .

This means,

$$\hat{C}_n : \mathcal{X} \to \{\text{a subset of } \mathcal{Y}\}$$

with $(X_{n+1}, Y_{n+1}) \sim P$, that respects

$$\mathbb{P}\big(Y_{n+1} \in \hat{C}_n(X_{n+1})\big) \geq 1 - \alpha$$

Let's assume that the random variables are independent and identically distributed $(i.i.d)$.The most trivial solution to achieve this goal is (3):

$$\hat{C}_n(X_{n+1}) = \begin{cases} \mathcal{Y} & \text{with probability } 1 - \alpha \\ \varnothing & \text{with probability } \alpha \end{cases}$$

In order not to have any kind of trivial solutions, we want the following property:

Property: Meaningful predictive set (4; 1; 3; 2)

The predictive set should be as small as theoretically possible. Implying that, the easier it is to predict Y_{n+1} from X_{n+1}, the smaller the set $\hat{C}_n(X_{n+1})$ should be, and the inverse should also be true.

We can achieve this constraint following the intuition that if we have a predictive model $\hat{f}_n$ outputting a value, we can use it to predict a set. To complete all the requirements described in the Introduction and the above paragraphs we still need few more properties.

Other properties (2)

- *The confidence intervals should be agnostic to the underlying predictive model.*
- *The elaboration of a confidence interval should be distribution-free.*
- *Validity should hold for finite samples ($n \leq \infty$), not just asymptotically (i. e. $n \to \infty$).*
- *Validity should hold for all models and data distribution.*

2.2 Theoretical framework for Conformal Prediction

In this part, we will focus on the theory in a regression setting. To lay down the main principles of conformal prediction we must first assume that our random variables $(X_i, Y_i) \sim P$ for $i = 1, \ldots, n$ are exchangeable, or *i.i.d.* Note that exchangeability is a weaker assumption than *i.i.d.*

Definition: Exchangeability (2)

The joint distribution of $Y_1, \ldots, Y_{n+1}$ doesn't change when permutted:

$$(Y_1, \ldots, Y_{n+1}) \overset{d}{=} (Y_{\sigma(1)}, \ldots, Y_{\sigma(n+1)})$$

for any permutation σ.

Focusing on the set $Y_i \in \mathbb{R}$ for $i = 1, \ldots, n$, the goal defined above is to find an interval C_n which satisfies the probability $\mathrm{P}(Y_{n+1} \leq \hat{q_n}) \geq 1 - \alpha$, where $\hat{q_n}$ is an empirical quantile. In other words, we want to create a confidence interval that has a $1-\alpha$ probability of containing the next point. First, we need a strong definition for quantiles:

Definition: Empirical quantile (2)

Given a set of point $Y_{(1)}, \ldots, Y_{(n)}$ sorted from lowest to largest, and $\alpha \in (0, 1)$. The empirical quantile is: $\hat{q_n}(Y_1, \ldots, Y_n) = Y_{(\lceil (1-\alpha) \times n \rceil)}$.

The key idea behind conformal prediction is to achieve finite-sample valid coverage by exploiting exchangeability.

The quantile lemma (2)

If $(Y_1, \ldots, Y_n, Y_{n+1})$ *are exchangeable, then for any* $\alpha \in (0,1)$*:*

$$\mathbb{P}\big(Y_{n+1} \leq \hat{q}_n(Y_1, \ldots, Y_n, Y_{n+1})\big) \geq 1 - \alpha.$$

Moreover, if $Y_1, \ldots, Y_n, Y_{n+1}$ *are almost surely distinct, then:*

$$\mathbb{P}\big(Y_{n+1} \leq \hat{q}_n(Y_1, \ldots, Y_n, Y_{n+1})\big) \leq 1 - \alpha + \frac{1}{n+1}.$$

Proof: The quantile lemma (3; 2)

In the simplest setting (without features), lets assume we have exchangeable data points $Y_1, \ldots, Y_n$, a natural goal is to construct a one-sided prediction interval of the form $\hat{C}_n = (-\infty, \hat{q}_n]$, such that,

$$\mathbb{P}\big(Y_{n+1} \leq \hat{q}_n\big) \geq 1 - \alpha.$$

While using the empirical $(1-\alpha)$ quantile of $Y_1, \ldots, Y_n$ only provides approximate coverage for finite n, conformal prediction achieves exact finite-sample validity by leveraging exchangeability. Indeed, if $Y_1, \ldots, Y_n, Y_{n+1}$ are exchangeable, then the rank of Y_{n+1} among $\{Y_1, \ldots, Y_n, Y_{n+1}\}$ is uniformly distributed over $\{1, \ldots, n+1\}$. Therefore,

$$\begin{aligned}
&\mathbb{P}\left(Y_{n+1} \text{ is among the } \lceil (1-\alpha)(n+1) \rceil \text{ smallest of } Y_1, \ldots, Y_n, Y_{n+1}\right) \\
&= \frac{\lceil (1-\alpha)(n+1) \rceil}{n+1} \geq 1 - \alpha.
\end{aligned}$$

This implies

$$\mathbb{P}\big(Y_{n+1} \leq Y_{(\lceil (1-\alpha)(n+1) \rceil)}\big) \geq 1 - \alpha,$$

where $Y_{(k)}$ denotes the k-th order statistic. Hence, defining

$$\hat{q}_n = Y_{(\lceil (1-\alpha)(n+1) \rceil)},$$

we obtain the desired finite-sample guarantee

$$\mathbb{P}\big(Y_{n+1} \leq \hat{q}_n\big) \geq 1 - \alpha.$$

Moreover, if the distribution is continuous (or almost surely distinct), then this result can be sharpened to

$$\mathbb{P}\big(Y_{n+1} \leq \hat{q}_n\big) = \frac{\lceil (1-\alpha)(n+1) \rceil}{n+1} \leq 1 - \alpha + \frac{1}{n+1}.$$

This concludes the proof.

The first section of the lemma, is a standard property of conformal inference (7). The second section was introduced by Jing Lei in (4).

However, a naive extension of this idea to predictive models fails. Given a predictor $\hat{f}_n$ trained on $(X_i, Y_i)_{i=1}^n$, we can define residuals

$$R_i = |Y_i - \hat{f}_n(X_i)|, \quad i = 1, \ldots, n,$$

Note that in this setting lower and upper bounds $(\hat{q}_n^l, \hat{q}_n^u)$ are determined using the same value $R_{(i)}$.Then we construct a prediction set of the form

$$\hat{C}_n(x) = \big[\hat{f}_n(x) - \hat{q}_n,\ \hat{f}_n(x) + \hat{q}_n\big],$$

where $\hat{q}_n$ is the $\lceil(1-\alpha)(n+1)\rceil$-th smallest value among $R_1, \ldots, R_n$. This construction would require

$$\mathbb{P}\big(R_{n+1} \leq \hat{q}_n\big) \geq 1 - \alpha, \quad \text{where} \quad R_{n+1} = |Y_{n+1} - \hat{f}_n(X_{n+1})|.$$

This fails because R_{n+1} is not exchangeable with $R_1, \ldots, R_n$: the model $\hat{f}_n$ has been trained on the data $(X_i, Y_i)_{i=1}^n$, making the training residuals artificially small, while the test residual is computed on an unseen point and is typically larger.

Example 1:

There is nothing more enlightening than an example. Let's assume we have a set of 2D random variables that are exchangeable, this set is presented in the following table:

x	5	3	7	3	2	7	8	1
y	0	3	9	5	4	6	8	6

Table 1: Dataset of (x, y) points, with n=8

We fit a linear model of the form:

$$\hat{f}_n(x) = \beta_0 + \beta_1 x$$

From the model, we obtain:

$$\beta_1 = 0.46875, \quad \beta_0 = 3.015625$$

Thus, the regression function is:

$$\hat{f}_n(x) = 3.015625 + 0.46875 \cdot x$$

The absolute residuals are:

$$R_i = |Y_i - \hat{f}_n(X_i)|$$

Then, we sort the absolute residuals:

$$R_{(1)} \leq R_{(2)} \leq \cdots \leq R_{(n)}$$

In our case:

$$[0.046875,\ 0.296875,\ 0.578125,\ 1.234375,\ 1.421875,\ 2.515625,\ 2.703125,\ 5.359375]$$

Let $\alpha = 0.25$. We compute the quantile index k:

$$k = \lceil (1-\alpha)\, n \rceil$$

With $n = 8$:

$$k = \lceil 0.75 \times 8 \rceil = 6$$

The quantile is:

$$\hat{q}_n = R_{(k)} = 2.515625$$

The conformal prediction interval is:

$$[\hat{f}_n(x) - \hat{q}_n,\ \hat{f}_n(x) + \hat{q}_n]$$

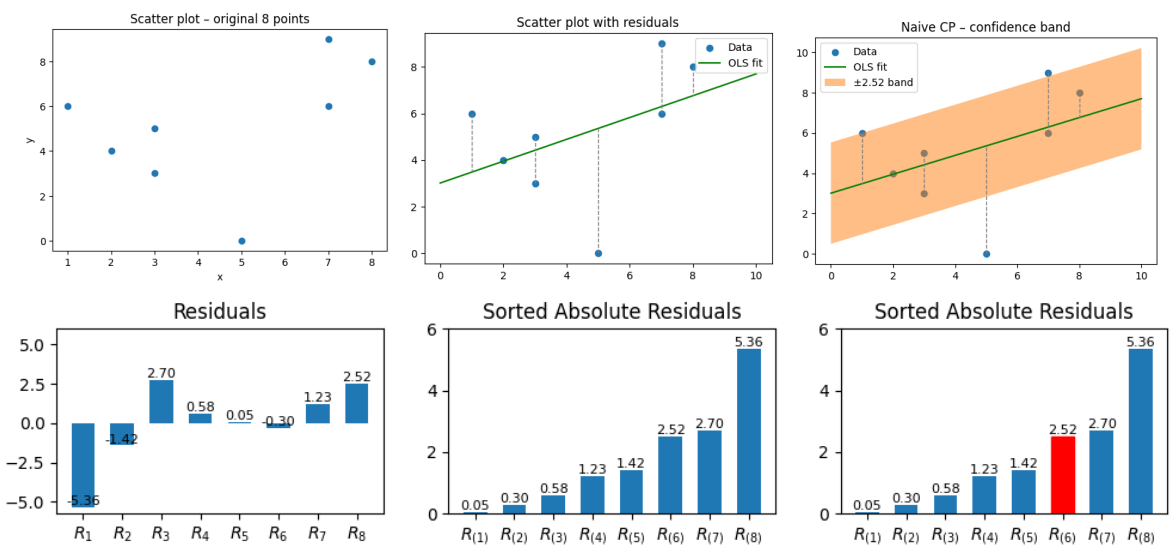

Figure 1: Empirical quantiles for confidence intervals based on a linear regression.

As previously mentioned, the guarantees works because the linear model and the confidence band were fitted on the same dataset. But, the goal of any predictive model is to predict the value of unseen data points. Here we generate 10 new data point in the same setting (exchangeable) as the previous ones, $X_{i=9}^{18}$.

x	6	7	6	2	7	3	7	5	1	5
y	3	4	9	6	4	7	2	4	7	1

Table 2: New dataset of (x, y) points

We predict the value using the fitted model on $X_{i=1}^{8}$. In the Figure 2, we see that on the 10 generated points only 4 are inside the confidence band, and 6 are outside. While the garantee

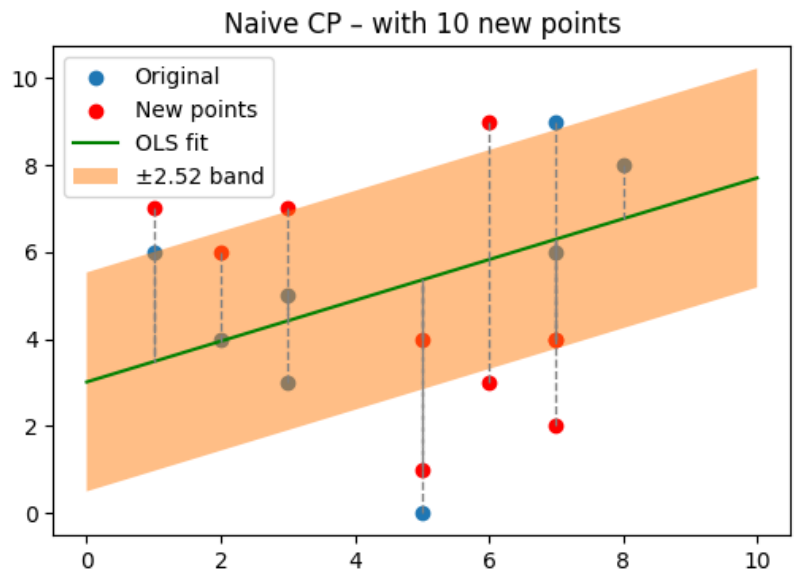

Figure 2: Empirical quantiles for confidence intervals based on a linear regression with 10 more points.

was to have $\mathbb{P}\big(R_{8+i} \leq \hat{q}_n\big) \geq 0.75$. Therefore, the probability of having $\mathbb{P}\big(R_{8+i} \leq \hat{q}_n\big) = 0.4$ is quite unlikely.

As shown previously, this naive method leads to undercoverage. This specific issue will be solved in chapters 3 and 4.

3 Split Conformal Prediction (SCP)

Split conformal prediction (SCP), is based on the idea of splitting the dataset to build the conformal set. We will define the predictive model as $\hat{f}_n$, it can be a regression (of any kind) or a classification model. Usually it is a machine learning or a statistical inference model based on supervised learning. This requires to split the dataset in two: Train and Test. In SCP, the dataset is split in three: Train 1 ($\mathcal{D}_1$), Train 2 ($\mathcal{D}_2$), and Test ($\mathcal{D}_3$). $\mathcal{D}_1$ is the classic training dataset, used as in every supervised learning task. $\mathcal{D}_2$ is referred to as the calibration set, on which we will derive the set $\hat{C}_n$. $\mathcal{D}_3$ is used as every Test set. Note that $\mathcal{D}_1 \cap \mathcal{D}_2 = \varnothing$ (3; 2).

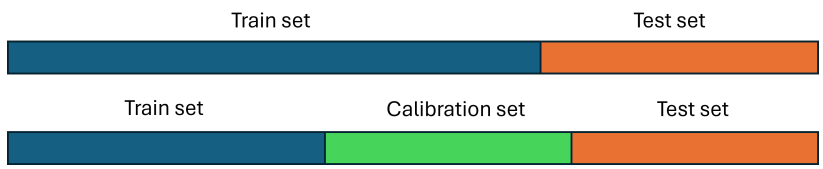

Figure 3: Dataset split in classical machine learning versus SCP

The splits between train ($\mathcal{D}_1$) and calibration ($\mathcal{D}_2$) can be of any proportion, but if one or the other is to small the underlying model $\hat{f}_n$ or the confidence band $[\hat{f}_n(x) - \hat{q}_n,\ \hat{f}_n(x) + \hat{q}_n]$ won't be very precise.

3.1 SCP in Regression

Defining $\hat{f}_n$ as the regression model, we want to predict the value $Y_{n+1} \in \mathcal{D}_3$ from (X_i, Y_i) for $i = 1, \ldots, n$ belonging to $\mathcal{D}_1$. From this point there is no difference with a classical regression setting. Now, we want to create the prediction band, which is the real focus of SCP. Recall that every supervised learning algorithm calculate a score to minimize (or maximize) in order to adjust the weights (parameters). For CP we will define such a score (similar as in Section 2) as $R_i = |Y_i - \hat{f}_n|$ for $i = 1, \ldots, n$. We will use this score to create our prediction band using $\mathcal{D}_2$. Note that we create our prediction band on $\mathcal{D}_2 \neq \mathcal{D}_1$ because, as we train on X_i for $i = 1, \ldots, n \in \mathcal{D}_1$ the regression adapted to X_i but not to X_{n+1}, therefore $R_{n+1} \in \mathcal{D}_3$ is stochastically bigger than R_i.

3.1.1 Mean Confomal Regression

Mean Conformal Regression is the simplest of all SCP methods (except the trivial ones). We train on $\mathcal{D}_1$, then we calculate a score $R_i = |Y_i - \hat{f}_{n_1}|$, $i = 1, \ldots, n_2$ for $i \in \mathcal{D}_2$. We order R_i from smallest to largest and we add the value infinity to the set: $\{R_{(1)}, \ldots, R_{(n_2)}\} \cap \infty$, this ensure that we take into consideration the worst case scenario. Indeed, if we take $1 - \alpha = 1$, the guarantee would hold only if we include the whole space $\mathcal{Y}$. We select a score in the set such that the error rate α is respected. This means, we select $R_{(i)}$ such that $R_{(i)} = R_{(\lceil (1-\alpha)(n+1) \rceil)} = \hat{q}_{n_2}$. Where $y \in \hat{C}_{n_2}(X)$ if $|y - \hat{f}_{n_1}| \leq \hat{q}_{n_2}$ (3; 2). Therefore the prediction band can be defined as:

$$\hat{C}_{n_2}(x) = [\hat{f}_{n_1} - \hat{q}_{n_2}, \hat{f}_{n_1} + \hat{q}_{n_2}]$$

This gives us to the following theorem,

Theorem: Split Conformal Prediction (4; 3; 2)

If (X_i, Y_i), $i = 1, \ldots, n$ are exchangeable and almost surely distinct, then for a new pair (X_{n+1}, Y_{n+1}) following the same constraint, we have

$$\mathbb{P}\big(Y_{n+1} \in \hat{C}_{n_2}(X_{n+1})\big) \geq 1 - \alpha.$$

For the conformal prediction band $\hat{C}_{n_2}$, if we assume additionally that for all $y \in \mathbb{R}$, the fitted absolute residuals

$$R_{y,i} = \left|Y_i - \hat{f}_{n_1}(X_i)\right|, \quad i = 1, \ldots, n_2$$

have a continuous joint distribution, then it also holds that

$$\mathbb{P}\big(Y_{n+1} \in \hat{C}_{n_2}(X_{n+1})\big) \leq 1 - \alpha + \frac{1}{n_2 + 1}.$$

Proof: Split Conformal Prediction theorem (4)

The lower bound follows directly by the exchangeability property between the residual at (X_{n+1}, Y_{n+1}) and those at (X_i, Y_i), $i \in \mathcal{D}_2$. For the upper bound, assuming a continuous joint distribution of residual, and hence no ties, the set $\hat{C}_{n_2}(X_{n+1})$ includes the values of Y_i such that $|Y_i - \hat{f}(X_{n+1})|$ is among the $\lceil (n_2 + 1)(1 - \alpha) \rceil$ smallest in $\{R_i : i \in \mathcal{D}_2\}$. Then again by exchangeability,

$$\mathbb{P}\big(Y_{n+1} \in \hat{C}_{n_2}(X_{n+1})\big) \leq \frac{\lceil (n_2 + 1)(1 - \alpha) \rceil}{n_2 + 1} \leq 1 - \alpha - \frac{1}{n_2 + 1}.$$

This concludes the proof.

In general, improving the quality of the predictive model $\hat{f}_n$ result is smaller conformal interval. Indeed, better models lead to smaller residuals (4).

We can also use the notation:

$$\mathrm{P}(Y_{n+1} \in \hat{C}_n(X_{n+1} | (X_i, Y_i) \in \mathcal{D}_1)) \in [1 - \alpha, 1 - \alpha + \frac{1}{n_2 + 1})$$

Which is equivalent to:

$$Y_{n+1} \in \hat{C}_{n_2}(X_{n+1}) \Leftrightarrow R_{n+1} \leq \hat{q}_{n_2} = R_{(\lceil (1-\alpha)\,(n_2+1) \rceil)}$$

The complete algorithm can be found in the appendix A.1.

Example 2:
Given the same dataset as in example 1:

$$\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^{18}$$

We split it into three disjoint subsets:

$$\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2 \cup \mathcal{D}_3$$

We train a linear regression model on the training set. Using the training data, we obtain:

$$\beta_0 = 2.4268, \quad \beta_1 = 0.4512$$

For each calibration point:

$$\hat{Y}_i = \hat{f}(X_i)$$

we compute absolute residuals on the calibration set and sorting them we obtain:

$$[1.5854,\ 3.1220,\ 3.2195,\ 3.4146,\ 3.6829,\ 4.1220,\ \infty]$$

Let $\alpha = 0.25$. The conformal quantile index is:

$$k = \lceil 0.75 \times 7 \rceil = 6$$

Thus:

$$q = |R|_{(k)} = 4.1220$$

For a new input x, the conformal prediction interval is:

$$C(x) = [\hat{f}(x) - q,\ \hat{f}(x) + q]$$

This interval guarantees marginal coverage:

$$\mathbb{P}(Y \in C(X)) \geq 1 - \alpha$$

We can confirm that this is the case in our example, as we can observe on Figure 5. Only one of the six test data points is outside the confidence band (1/6 instead of 6/10 previously).

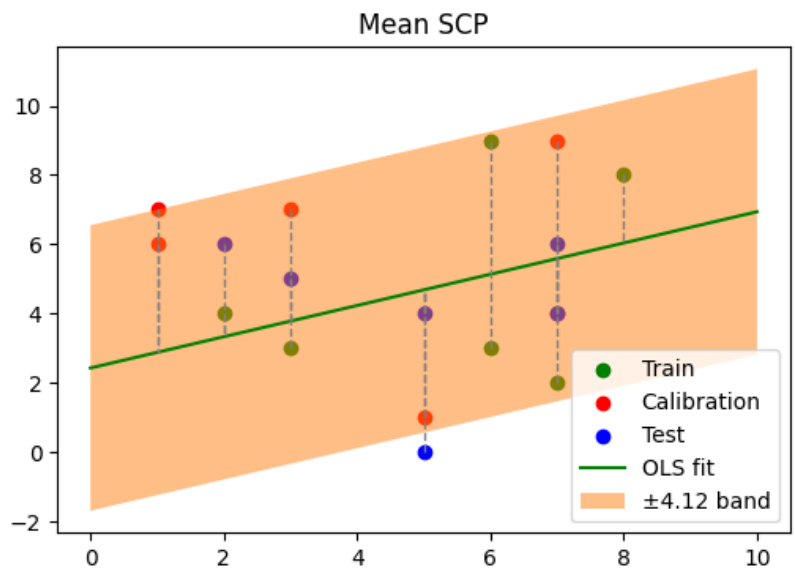

Figure 4: Mean SCP with 18 data points (6 train [green], 6 calibration [red], 6 test [blue]).

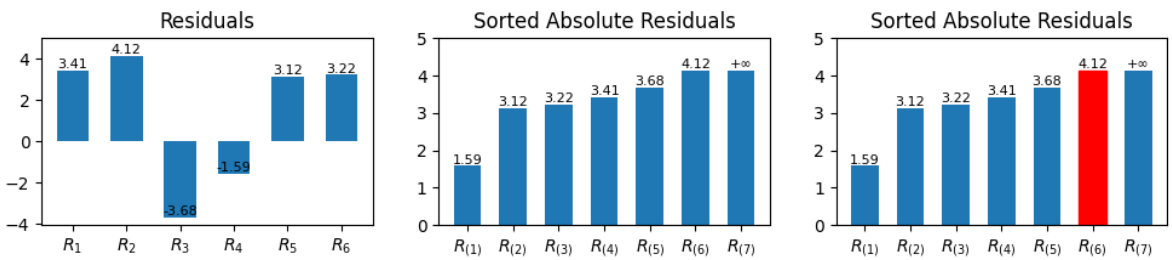

Figure 5: Finding the SCP quantile.

Nevertheless, the main disadvantage of this method is that the bandwidth isn't adaptive. Indeed, we may want to have a smaller bandwidth if we are more confident in predicting value with smaller variance, and a larger bandwidth when we want to predict value with a larger variance.

3.1.2 Conformalized Quantile Regression

Quantile regression is a statistical tool that produces a regression to predict the values of a given quantile (4; 3; 2). In conformalized quantile regression (CQR), we first fit two quantile regressions: $\hat{f}_{n_1}^{\alpha/2}(x)$ and $\hat{f}_{n_1}^{1-\alpha/2}(x)$ (lower and upper regressions) based on the error level α, on $(X_i, Y_i), i \in \mathcal{D}_1$. We compute calibration set scores on $\mathcal{D}_2$:

$$R_i = max\{\hat{f}_{n_1}^{\alpha/2}(X_i) - Y_i, Y_i - \hat{f}_{n_1}^{1-\alpha/2}(X_i)\},\ i \in \mathcal{D}_2$$

As previously, $R_{(\lceil(1-\alpha)(n_2+1)\rceil)} = \hat{q}_{n_2}$. Then the CQR set is:

$$\hat{C}_n(x) = [\hat{f}_{n_1}^{\alpha/2}(x) - \hat{q}_{n_2}, \hat{f}_{n_1}^{1-\alpha/2}(x) + \hat{q}_{n_2}]$$

The full algorithm can be found in the appendix A.2.

Example 3:
Using the same setting as in example 1 and 2, we obtain the Figure 6:

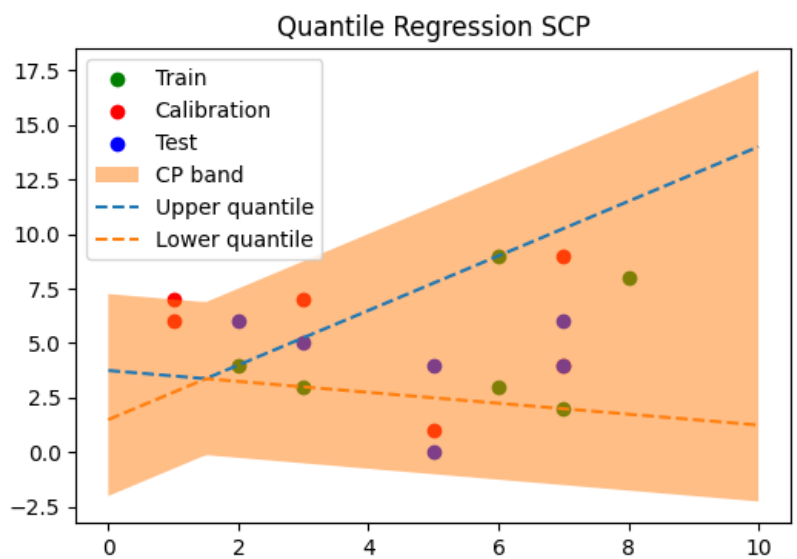

Figure 6: Quantile SCP with 18 data points (6 train [green], 6 calibration [red], 6 test [blue]).

3.1.3 Weighted SCP

Again, to account for heteroskedasticity, we can extend the standard conformal prediction to a locally weighted version (4; 3; 2). Instead of using absolute residuals alone, we normalize them by an approximation of the local error spread. Let $\hat{f}_n$ be our predictive model (mean estimation) and $\hat{\rho}_n$ an estimate of the conditional mean absolute deviation (MAD) of $Y - \hat{f}_{n_1}(X)$ given $X = x$. In practice it is also a supervised predictive model (See GitHub for more practical details). The weighted (sometime referred as studentized) residuals are defined as:

$$R_i = \frac{|Y_i - \hat{f}_{n_1}(X_i)|}{\hat{\rho}_{n_1}(X_i)}, \quad i \in \mathcal{D}_2.$$

For a new point x, the corresponding conformity score is:

$$R_{n+1} = \frac{|y - \hat{f}_{n_1}(x)|}{\hat{\rho}_{n_1}(x)}.$$

The functions $\hat{f}_{n_1}(x)$ and $\hat{\rho}_{n_1}(x)$ are estimated using the training set $\mathcal{D}_1$. Let $\alpha \in (0,1)$ again the the quantile is define as:

$$\hat{q}_{n_2} = R_{(\lceil (1-\alpha)(n_2+1) \rceil)}$$

the $\lceil (1-\alpha)(n_2+1) \rceil$-th smallest value among $\{R_i : i \in \mathcal{D}_2\}$. The weighted conformal prediction interval at a new point x is given by:

$$C(x) = [\, \hat{f}_{n_1}(x) - \hat{\rho}_{n_1}(x) \times \hat{q}_{n_2},\ \hat{f}_{n_1}(x) + \hat{\rho}_{n_1}(x) \times \hat{q}_{n_2} \,].$$

Using locally weighted residuals preserves the theoretical guarantees of conformal prediction. But, it improve adaptivity under heteroscedasticity.

The complete algorithm can be found in the appendix A.3.

Example 4:
Using the same setting as in the previous example, but with a different shuffle split, as the previous split was less relevant for the task. We obtain the Figure 7:

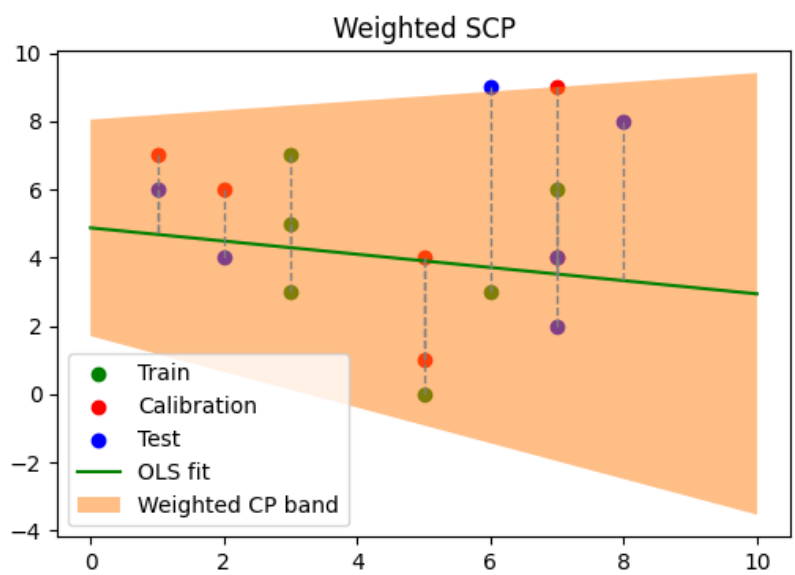

Figure 7: Weighted SCP with 18 data points (6 train [green], 6 calibration [red], 6 test [blue]).

3.2 SCP in Classification

3.2.1 Vanilla Split Conformal Classification

SCP in Classification or Conformal Classification takes root in classification tasks, based on any supervised classification model (3; 2). Therefore we fit a classifier $\hat{f}_n$ to $(X_i, Y_i), i \in \mathcal{D}_1$.

The probability or likelihood score of x to be classified as class k, among the $k = 1, \ldots, j$ classes can be any metric that is positively oriented (the higher, the more likely it is). We define the likelihood to be classified as k to be $\hat{f}_n(x; k)$. We note R_i as the likelihood score: $R_i = \hat{f}_n(X_i, Y_i), i \in \mathcal{D}_2$. It is important to notice that we don't use the most likely output, but the likelihood of the true class Y_i. In this method we rank the scores from lowest to highest: $\{R_{(1)}, \ldots, R_{(n)}\} \cap 1$. We define $\hat{q}_n = R_{(\lfloor \alpha(n+1) \rfloor)}$ among $\mathcal{D}_2$. This creates the following set,

$$\hat{C}_n(x) = \{k : \hat{f}_n(x; k) \geq \hat{q}_n\}$$

Note that the previous guarantees hold (3). Another popular version of this method exist and is based on a conformal score that is defined as $R_i = 1 - \hat{f}_n(X_i, Y_i), i \in \mathcal{D}_2$ (3).

The complete algorithm can be found in the appendix A.4.

Example 5:

We consider a multiclass classification problem on the MNIST dataset, with labels $\mathcal{Y} = \{0, 1, \ldots, 9\}$. The predictive model $\hat{f}_n$ is a convolutional neural network trained on 20% of the MNIST training set. The model architecture is:

$$\texttt{Conv2d}(1, 1,\ \text{kernel size} = 3,\ \text{padding} = 1) \rightarrow \texttt{ReLU} \rightarrow \texttt{MaxPool2d}(4) \rightarrow \texttt{Flatten}$$
$$\rightarrow \texttt{Linear}(1 \times 7 \times 7, 14) \rightarrow \texttt{ReLU} \rightarrow \texttt{Linear}(14, 10)$$

The network is trained for one epoch using Adam with a learning rate of 0.001. The test accuracy obtained is 0.6498. It is intentionally low for more meaningfull results. We perform the conformal classification on 14 test images (Figure 8), where the softmax outputs are in the table 3. The first 10 examples are used as a calibration set, and the last 4 examples are used as the test set.
For calibration points (X_i, Y_i), we define the conformal score as $R_i = \hat{f}_n(X_i, Y_i)$. After sorting and and adding 1, We obtain the following set:

$$[0.1420,\ 0.1688,\ 0.2072,\ 0.3449,\ 0.3822,\ 0.4669,\ 0.5097,\ 0.5525,\ 0.6676,\ 0.6942,\ 1]$$

With $\alpha = 0.25$, we have $q = 0.1688$. By defining $\hat{C}_n(x) = \{k : \hat{f}_n(x; k) \geq \hat{q}_n\}$, we obtain the confidence band:

$$\widehat{C}(x_{11}) = \{2, 3\}$$
$$\widehat{C}(x_{12}) = \{4, 9\}$$
$$\widehat{C}(x_{13}) = \{0\}$$
$$\widehat{C}(x_{14}) = \{0, 2\}$$

While true labels are $(2, 9, 0, 0)$. All are contained in their prediction sets, so the empirical coverage is 100%.

idx	true	pred	0	1	2	3	4	5	6	7	8	9
1	5	4	0.0220	0.0185	0.0071	0.0096	0.3079	0.1420	0.0235	0.0522	0.1442	0.2731
2	6	6	0.1103	0.0050	0.2545	0.0465	0.0126	0.0233	0.4669	0.0061	0.0614	0.0134
3	7	7	0.0112	0.0423	0.0044	0.0690	0.0344	0.0304	0.0046	0.6942	0.0379	0.0714
4	2	2	0.0653	0.0240	0.3822	0.3772	0.0036	0.0254	0.0414	0.0393	0.0372	0.0045
5	0	0	0.3449	0.0028	0.0648	0.1716	0.0281	0.1549	0.0284	0.0721	0.0890	0.0435
6	1	1	0.0031	0.6676	0.0700	0.0346	0.0285	0.0188	0.0892	0.0255	0.0373	0.0255
7	8	4	0.0346	0.0965	0.0450	0.0602	0.1844	0.1030	0.0664	0.1056	0.1688	0.1355
8	9	7	0.0420	0.0208	0.0060	0.0508	0.1497	0.0995	0.0118	0.3160	0.0961	0.2072
9	2	2	0.0248	0.0266	0.5097	0.3265	0.0011	0.0108	0.0596	0.0198	0.0194	0.0016
10	3	3	0.0325	0.0424	0.0830	0.5525	0.0112	0.0399	0.0227	0.1545	0.0458	0.0156
11	2	3	0.1037	0.0079	0.3445	0.3814	0.0039	0.0262	0.0390	0.0406	0.0489	0.0040
12	9	9	0.0113	0.0873	0.0048	0.0205	0.2342	0.0804	0.0204	0.1680	0.0749	0.2981
13	0	0	0.3677	0.0018	0.1553	0.0673	0.0271	0.1524	0.0690	0.0144	0.1155	0.0294
14	0	2	0.2747	0.0017	0.4052	0.0577	0.0045	0.0724	0.1243	0.0051	0.0469	0.0076

Table 3: Predicted class probabilities for 14 MNIST examples. The first 10 examples form the calibration set used to compute the conformal threshold, while the last 4 examples form the test set on which prediction sets are constructed.

3.2.2 Adaptive Regression Sets (APS)

The authors in the paper (5), proposed another approach using cumulative likelihood, which is often referred as Adaptive Regression Sets (APS). First, we sort each likelihood score:

$$\hat{f}_n(X_i; k_{(1)_+}) \geq \hat{f}_n(X_i; k_{(2)_+}) \geq \cdots \geq \hat{f}_n(X_i; k_{(c)_+})$$

For all the c classes, where $k_{(i)_+}$ denotes the i-th biggest value of k (3; 1). We define the conformity score as:

$$R_i = \sum_{j=1}^{t} \hat{f}_n(X_i; k_{(j)_+}) \qquad \text{where } k_{(t)_+} = Y_i \text{ for } i \in \mathcal{D}_2$$

R_i is the cumulative score of all the scores from largest to smallest, up to the score associated with the true class. We have $\hat{q}_n = R_{(\lceil (1-\alpha)(n+1) \rceil)}$ for $i \in \mathcal{D}_2$.

The conformal set is:

$$\hat{C}_n(x) = \{k_{(1)_+}, \ldots, k_{(t)_+}\}$$

With,

$$(t) = min\{t : \sum_{j=1}^{t} \hat{f}_n(x; k_{(j)_+}) \leq \hat{q}_n)\}$$

Finally, note that some researchers presented a regularized version of APS called RAPS (6), that produce better sets in practice.

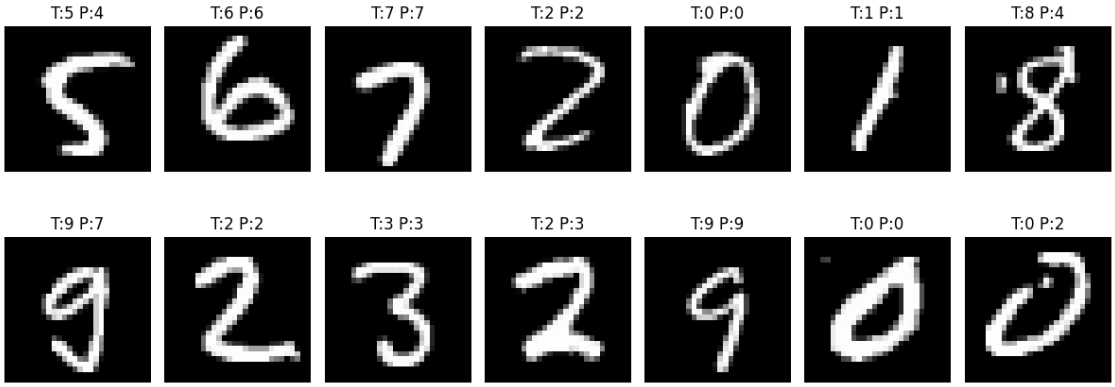

Figure 8: The 14 input data for calibration set and and test set, ordered from 1 to 14.

The complete algorithm can be found in the appendix A.5.

Example 6:

For this example we keep the exact same setting as for example 5, but we adapt it to APS. For calibration points (X_i, Y_i), we first sort the predicted probabilities in decreasing order, $\hat{f}_n(X_i; k_{(1)_+}) \geq \hat{f}_n(X_i; k_{(2)_+}) \geq \cdots \geq \hat{f}_n(X_i; k_{(10)_+})$. We then define the APS conformity score as the cumulative probability up to the true label, $R_i = \sum_{j=1}^{t_i} \hat{f}_n(X_i; k_{(j)_+})$, where $k_{(t_i)} = y_i$. After computing these cumulative scores and sorting them, we obtain:

$$[0.3449,\ 0.3533,\ 0.3822,\ 0.4669,\ 0.5097,\ 0.5232,\ 0.5525,\ 0.6676,\ 0.6942,\ 0.8672]$$

With $\alpha = 0.25$, we have $\hat{q}_n = 0.6942$. The conformal set is defined as $\widehat{C}_n(x) = \{k_{(1)_+}, \ldots, k_{(t)_+}\}$, where

$$t = \min\left\{ t : \sum_{j=1}^{t} \hat{f}_n(x; k_{(j)_+}) \geq \hat{q}_n \right\}.$$

Applying this rule, we obtain the prediction sets:

$$\begin{aligned}
\widehat{C}(x_{11}) &= \{3, 2\} \\
\widehat{C}(x_{12}) &= \{9, 4, 7\} \\
\widehat{C}(x_{13}) &= \{0, 2, 5, 8\} \\
\widehat{C}(x_{14}) &= \{2, 0, 6\}
\end{aligned}$$

The true labels are $(2, 9, 0, 0)$. Again, all are contained in their prediction sets, so the empirical coverage is 100%.

4 Full Conformal Prediction (FCP)

Full Conformal Prediction is a family of Conformal prediction in which we want similar guarantees as in SCP, but we want to get rid of the Calibration set (4; 3; 1; 2). We have $X_{n+1} \in \mathcal{X}$ and we want to know for every $y \in \mathcal{Y}$ associated with x if $y \in \hat{C}_n(x)$.
We train $\hat{f}$ on $(X_1, Y_1), \ldots, (X_n, Y_n), (X_{n+1}, y)$, the augmented training set with $n+1$ points. Therefore, the point predictor is $\hat{f}_{n,(X_{n+1},y)}$, and we define the conformal score as:

$$R_i = |Y_i - \hat{f}_{n,(X_{n+1},y)}(X_i)| \qquad i = 1, \ldots, n$$

$$R_{n+1} = |y - \hat{f}_{n,(X_{n+1},y)}(X_{n+1})|$$

The conformal set becomes,

$$\hat{C}_n(x) = \{y \in \mathcal{Y} : R_{n+1} \leq \hat{q}_n\}$$

With $\hat{q}_n = R_{(\lceil (1-\alpha)(n+1) \rceil)}$ from the increasing ranked conformal score $R_{(1)}, \ldots, R_{(n)}, R_{(n+1)}$. Note that, we fit $\hat{f}_{n,(X_{n+1},y)}$ on $(X_1, Y_1), \ldots, (X_n, Y_n), (X_{n+1}, y)$, we set the confidence interval using only the residuals of $(X_1, Y_1), \ldots, (X_n, Y_n)$.

Using the different tools we laid out, we can now show the following theorem:

Theorem: Full Conformal Prediction (4; 3; 2)

If (X_i, Y_i), $i = 1, \ldots, n$ are exchangeable and almost surely distinct, then for a new pair (X_{n+1}, Y_{n+1}) following the same constraint, we have

$$\mathbb{P}\big(Y_{n+1} \in \hat{C}_n(X_{n+1})\big) \geq 1 - \alpha.$$

For the conformal prediction band $\hat{C}_n$, if we assume additionally that for all $y \in \mathbb{R}$, the fitted absolute residuals

$$R_i = \left| Y_i - \hat{f}_{n,(X_i,y)}(X_i) \right|, \quad i = 1, \ldots, n$$

have a continuous joint distribution, then it also holds that

$$\mathbb{P}\big(Y_{n+1} \in \hat{C}_n(X_{n+1})\big) \leq 1 - \alpha + \frac{1}{n+1}.$$

Proof: Full Conformal Prediction theorem (4; 2)

For this proof we need the exchangeable score lemma and the symmetrical algorithm definition provided below. The lower bound comes from the definition of the conformal interval. For the upper bound, by assuming a continuous joint distribution of the fitted residuals, we know that the values $R_1, \ldots, R_{n+1}$ are all distinct with probability one. Then, by exchangeability (exchangeable scores lemma), the set C_n in is equivalent to the set of all points y such that R_{n+1} ranks among the $\lceil (n+1)(1-\alpha) \rceil$ smallest of all $R_1, \ldots, R_{n+1}$. Therefore,

$$\mathbb{P}\big(Y_{n+1} \in C_n(X_{n+1})\big) \leq \frac{\lceil (n+1)(1-\alpha) \rceil}{n+1} \leq 1 - \alpha + \frac{1}{n+1}.$$

This concludes the proof.

Note that, even if we aim for model-free prediction, here we need $\hat{f}_n$ to be a symmetric function of the data points (because we refit the model each time, unlike in SCP) (4).

Definition: Symmetrical algorithm (2)

A deterministic algorithm $f : (U_1, \ldots, U_n) \mapsto \hat{f}$ *is* symmetric *if for any permutation* σ *of* $[1, n]$*:*

$$f(U_1, \ldots, U_n) \overset{a.s.}{=} f(U_{\sigma(1)}, \ldots, U_{\sigma(n)}).$$

Lemma: Exchangeable scores (2)

If the algorithm $f : (U_1, \ldots, U_n) \mapsto \hat{f}$ *is* symmetric*, and* $(X_i, Y_i)_{i=1}^{n+1}$ *are exchangeable, then* $R_1, \ldots, R_{n+1}$ *are exchangeable, with*

$$R_i := \left| Y_i - \hat{f}_{n+1}(X_i) \right|.$$

Again, the proof is similar to the one of the quantile lemma. This theorem confirm valid finite-sample coverage, as well as the fact that the method doesn't substantially over-cover (4).

FCP method require to do the operation for each $y \in \mathbb{R}$ for a given x, so the conformal set consists of an infinite set of segments on $\mathcal{Y}$ over the domain of $f : dom(f)$. This allows to have the following guarantee:

$$\mathrm{P}(Y_{n+1} \in \hat{C}_n(X_{n+1}) \in [1-\alpha, 1-\alpha+\frac{1}{n+1})$$

without any calibration set.

The complete algorithm can be found in the appendix A.6.

In practice, this isn't possible, so we have to define a subset of $\mathcal{Y}$ and $\mathcal{X}$ we want to test. Nonetheless, FCP is a very expensive method computationally, and also more complex than SCP. The main advantage is to be able to train on the full dataset without having to split it, which is great when the amount of data is limited. FCP works exactly the same as SCP, whcich means that we can perform regression and classification task, as well as weighted CP and CQR, just like in SCP.

To better grasp the concept of FCP, here is an example of a regression task. Note that using Deep Learning in the context of FCP isn't appropriate due to the high computational cost. In Deep Learning the dataset often consist of high dimensional tensor, therefore computing multiple run per input for multiple inputs is too expensive to be realistic (even if we test a subset of $\mathcal{X}$, and $\mathcal{Y}$). SCP is better suited for this kind of setting.

Example 7:
In this regression example, we use the same dataset as before using Full Conformal Prediction. First, we test each candidate point (i,j) where i and j are integers $\in [0, 9]$. We temporarily add the point to the dataset, retrain a linear regression model, and compute residual errors on the original data to get the quantile. Then, we check if the candidate point's error is below the quantile threshold. If yes, it is added to the set. Figure 12 present the results.

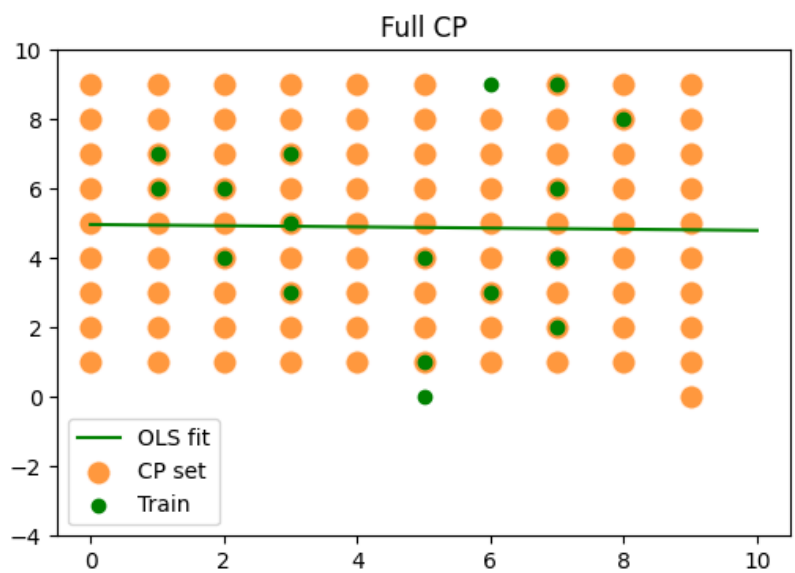

Figure 9: Full CP with 18 data points.

4.1 FCP as a Hypothesis test

Full conformal prediction (FCP) can be interpreted as a hypothesis test (7; 4). For each y, we consider the null hypothesis

$$H_0 : \; Y_{n+1} = y,$$

Note that the p-value is the probability, assuming H_0 is true, of obtaining a test statistic as extreme as or more extreme than the observed one. First, we augment the sample with (x, y), and fit the regression estimator $\hat{f}_{n,(x,y)}$ on the enlarged dataset. Then we define the augmented residuals, and compute

$$\pi(y) = \frac{1}{n+1} \sum_{i=1}^{n+1} \mathbf{1}\{R_i \leq R_{n+1}\} = \frac{1}{n+1} + \frac{1}{n+1} \sum_{i=1}^{n} \mathbf{1}\{R_i \leq R_{n+1}\},$$

where $\mathbf{1}$ is the indicator function. The above equation is the proportion of points in the augmented dataset whose fitted residual is smaller or equal to the test residual R_{n+1} (4). This implies that we can write the following equation,

$$\frac{1}{n+1} \sum_{i=1}^{n+1} \mathbf{1}\{R_i \leq R_{n+1}\} = 1 - \frac{1}{n+1} \sum_{i=1}^{n+1} \mathbf{1}\{R_i > R_{n+1}\} = 1 - \frac{1}{n+1} \sum_{i=1}^{n} \mathbf{1}\{R_i > R_{n+1}\}$$

Therefore, we can write

$$p(y) = 1 - \pi(y) = \frac{1}{n+1} \sum_{i=1}^{n} \mathbf{1}\{R_i > R_{n+1}\}$$

In this case, $p(y)$ is a conservative p-value for testing H_0. Under the null hypothesis, exchangeability implies that $\pi(y)$ is uniformly distributed on $\{1/(n+1), \ldots, 1\}$ and that,

$$\mathbb{P}(p(y) \leq \alpha) \leq \alpha.$$

The error rate is α, so we reject the null hypothesis in favor of the alternative hypothesis, if and only if $p(y) \leq \alpha$. Therefore, FCP is a hypothesis test whose confidence region corresponds to the $y \in \mathcal{Y}$ where the null hypothesis isn't rejected.

5 Challenges and Solutions in Conformal Prediction

5.1 Conditional Coverage

Conformal prediction is a framework for uncertainty quantification that produces prediction sets $C(X)$ such that:

$$P(Y \in C(X)) \geq 1 - \alpha,$$

for a specified miscoverage level $\alpha \in (0, 1)$ (4; 1; 2; 3). A key property of conformal prediction is that this guarantee holds without assumptions on the data distribution, for any predictive model, in finite samples.

However, this guarantee is marginal, meaning it holds on average over the data distribution. The standard conformal guarantee is taken over the randomness of the training and calibration data, and the randomness of the test point (X, Y). Thus, it ensures average correctness across all inputs. A stronger notion is conditional coverage:

$$P(Y \in C(X) \mid X = x) \geq 1 - \alpha \quad \forall x.$$

This requires that prediction sets achieve the target coverage for every input x. Marginal coverage is less interesting than conditional coverage. The authors in (1), give a meaningful example: Suppose 90% of individuals belong to group A and 10% to group B. If prediction sets always cover Y for group A but never for group B, the overall coverage is 90%. However, this violates conditional coverage, which requires at least 90% coverage within each group, not just on average.

Unfortunately, a fundamental result in conformal prediction theory states that under minimal assumptions (such as exchangeability), no method can achieve exact conditional coverage for all x with finite-length prediction intervals (4). To counter this fact, some research has been achieved to guarantee asymptotic conditional coverage. It has been shown that, under certain conditions, FCP and SCP have asymptotic conditional coverage (4).

5.2 Beyond Exchangeability

Conformal coverage guarantee lies in the fact that, the the data are excheangeable, or in the case of SCP, that the calibration and test set are excheangeable. In practice, this is rarely the case due to distribution shift, correlations, or other phenomena (8). In the following subsections we present two ways two deal with some specific kind of shifts in the underlying distribution, to recover some of the guarantee offered by CP.

5.2.1 Covariate Shift

Conformal prediction provides distribution-free prediction intervals with guaranteed marginal coverage, relying solely on the exchangeability. In practice, however, the training and test covariate distributions frequently differ, a setting called covariate shift. For example, the training dataset for predicting Y, comes from a sample of 20% from the class c_1 and 80% from the class c_2, but the proportions of the test set are 50% each. This is a covariate shift

for the feature c_i. Formally, training points $(X_i, Y_i) \overset{\text{i.i.d.}}{\sim} P_X \times P_{Y|X}$ for $i = 1, \ldots, n$, while the test point satisfies $(X_{n+1}, Y_{n+1}) \sim \tilde{P}_X \times P_{Y|X}$, with $\tilde{P}_X \neq P_X$ but the same conditional $P_{Y|X}$. Because the $n+1$ points are no longer exchangeable, standard conformal prediction loses its coverage guarantee and can substantially undercover. Indeed, the residuals of the test dataset come from a different region of the covariate space than most training residuals (9).

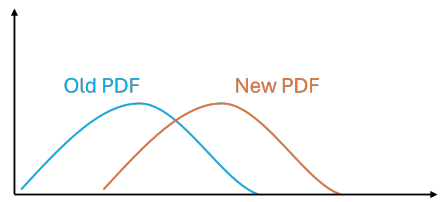

Figure 10: Example of covariate shift for the PDF of a feature

The natural solution is to reweight the distribution of the calibration scores R_i for $i = 1...n$, by the likelihood ratio (10),

$$w(x) = d\tilde{P}_X(x)/dP_X(x)$$

This is the ratio of the probability distribution function (PDF) of the shifted distribution over the original PDF (1). As we can deduce from Figure 11, when X_i is likely under the new distribution, it get a high weight $w(x)$, and inversely if it is unlikely it get a small weight.

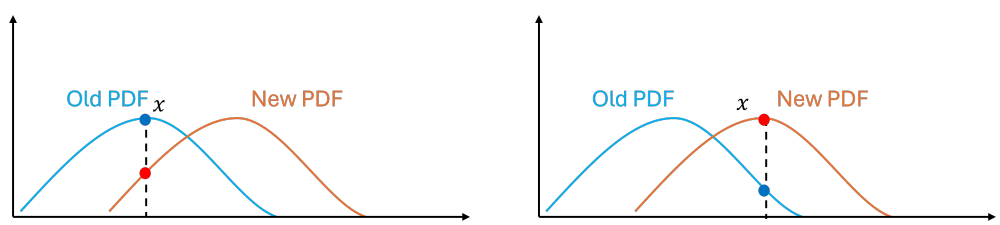

Figure 11: Examples of changing likelihood ratios

Therefore, we replace the uniform empirical distribution of conformal scores with the weighted distribution. Previously, each R_i had the same probability mass of $1/(n+1)$, therefore we had,

$$\frac{\lceil (1-\alpha)(n+1) \rceil}{n+1} = \frac{1}{n+1} \sum_{i=1}^{n} \mathbf{1}\{R_i \leq \hat{q}_{\text{unweighted}}\} + \frac{1}{n+1} \mathbf{1}\{\infty \leq \hat{q}_{\text{unweighted}}\},$$

with $\hat{q}_{\text{unweighted}} = R_{(\lceil (1-\alpha)(n+1) \rceil)}$ Under covariate shift, this becomes

$$\sum_{i=1}^{n} p_i^w(X_{n+1})\, \mathbf{1}\{R_i \leq \hat{q}_{\text{weighted}}\} + p_{n+1}^w(X_{n+1})\, \mathbf{1}\{\infty \leq \hat{q}_{\text{weighted}}\},$$

where $\hat{q}_{\text{weighted}}$ is the $(1-\alpha)$-quantile of the weighted distribution, i.e.,

$$\hat{q}_{\text{weighted}} = \inf\left\{ R_* : \sum_{i=1}^{n} p_i^w(X_{n+1})\,\mathbf{1}\{R_i \le R_*\} + p_{n+1}^w(X_{n+1})\,\mathbf{1}\{\infty \le R_*\} \ge 1-\alpha \right\}.$$

The weights for $i = 1, \ldots, n$ and for $i = n+1$ are defined by

$$p_i^w(X_{n+1}) = \frac{w(X_i)}{\sum_{j=1}^{n} w(X_j) + w(X_{n+1})}, \qquad \text{and} \quad p_{n+1}^w(X_{n+1}) = \frac{w(X_{n+1})}{\sum_{j=1}^{n} w(X_j) + w(X_{n+1})}.$$

First, note that this term $\frac{1}{n+1}\mathbf{1}\{\infty \le \hat{q}_{unweighted}\}$ account for the $+\infty$ that is added during the computation of the calibration score. It is multiplied by $p_{n+1}^w(X_{n+1})$, because $+\infty$ always accounts for the value X_{n+1}, as a proxy for the worst case scenario. From this technique, we recover the CP coverage guarantee for the lower bound.

When w is unknown, the authors propose estimating it by training a classifier on the calibration and test covariates (with labels $C_i = 0$ for training, $C_i = 1$ for test) and sets $\hat{w}(x) = \hat{p}(x)/(1-\hat{p}(x))$, where $\hat{p}(x)$ estimates $P(C = 1 \mid X = x)$. They experimented with logistic regression and random forests, and both recover near-nominal coverage on the experiment. Nonetheless, in low-dimensional settings w can be estimated from unlabeled test covariates, but the estimation is noisy, and in high-dimension this approach can be even more noisy.

5.2.2 Distribution Drift

For more general distribution drift (8) allow arbitrary fixed, pre-specified weights $w_i \in [0,1]$ encoding prior belief about which calibration points are representative of the test distribution. Other than that, it work exactly as the method described for the covariate shift. The calibration data $\{(X_i, Y_i)\}_{i=1}^n$ are drawn independently from potentially different distributions $\{P_i\}_{i=1}^m$, and the test point from P_{test}. The normalised weights

$$\tilde{w}_i = \frac{w_i}{w_1 + \cdots + w_n + 1}$$

are used to form a weighted quantile of conformity scores, and prediction sets are constructed in the standard way $C(x_i) = \{y_i : R_i \le \hat{q}_{\text{weighted}}\}$, with

$$\hat{q}_{\text{weighted}} = \inf\left\{ R_* : \sum_{i=1}^{n+1} \tilde{w}_i\,\mathbf{1}\{s_i \le R_*\} \ge 1-\alpha \right\}.$$

Using this method we still have some coverage guarantee (8). Let $\varepsilon_i = d_{\text{TV}}\big((X_i, Y_i),\,(X_{n+1}, Y_{n+1})\big)$ measure how much the i-th calibration distribution differs from the test distribution. Then

$$P\{Y_{n+1} \in C(X_{n+1})\} \;\ge\; 1 - \alpha - 2\sum_{i=1}^{n} \tilde{w}_i\,\varepsilon_i.$$

When $\tilde{w}_i\,\varepsilon_i$ is small, this calibration point contributes little to the coverage loss (8). In general, it can be small either because the distribution has not shifted much ($\varepsilon_i \approx 0$) or

because the point has been down-weighted ($\tilde{w}_i \approx 0$). Setting $\varepsilon_i = 0$ recovers exact $1 - \alpha$ coverage, so non-uniform weights never hurt in the i.i.d. paradigm. Note that if we have information about the drift we can dynamically change the weights, using weight decay for example. This is useful when we are operating in an online setting.

5.3 Mondrian Conformal Prediction

One of the challenges with standard conformal predictors, is that it only guarantee validity on average across all examples. Indeed, the error rate can significantly exceed the nominal error level α for certain categories or classes in our dataset. Mondrian Conformal Prediction (MCP) (7) address this by dividing examples into categories. The authors presented the Mondrian taxonomy, which is a measurable function k that maps each (index, example) pair to a category, such that each category can be interpreted as a rectangle in the index $\times$ example space.

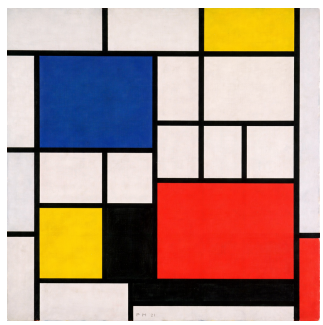

Figure 12: Mondrian inspired painting, generated by AI.

To put it simply, we compute Conformal Prediction separately within each identified category, ensuring conditional (category-wise) validity rather than just marginal validity. Therefore, we can go from,

$$\mathbb{P}\big(Y_{n+1} \in C(X_{n+1})\big) \geq 1 - \alpha$$

to,

$$\mathbb{P}\big(Y_{n+1} \in C(X_{n+1}) \,\big|\, K(X_{n+1}, Y_{n+1}) = c_j\big) \geq 1 - \alpha$$

where $K(X_{n+1}, Y_{n+1})$ maps (X_{n+1}, Y_{n+1}) to a categorical value $k = \{c_1, ..., c_m\}$ for any $j = 1, ..., m$.

In the following subsections we present the two principal applications of MCP, which are, group-balanced Conformal Prediction, and class-conditional Conformal Prediction.

5.3.1 Group-Balanced Conformal Prediction

In group-balanced conformal prediction we try to solve the necessity of having the lower bound coverage guarantee for any input $X = (x_1, ..., x_s)$ where x_i is a categorical value

$k = \{c_1, ..., c_m\}$, while so far it is only in average over all the test inputs. Some applications of this, would be to ensure fairness in coverage for a set of different categories (sex, ethnicity...). So, we would like to ensure the following gurantee,

$$\mathbb{P}\big(Y_{n+1} \in C(X_{n+1}) \,\big|\, x_i^{n+1} = c_j\big) \geq 1 - \alpha$$

for any $j = 1, ..., m$. To achieve this goal, the simplest technique is presented by Vovk (7). We treat each m class in k as a separate calibration set, and apply the conformal algorithm m times, to obtain $\hat{q}_1, ..., \hat{q}_m$ different threshold quantiles. Therefore, the conformal set becomes,

$$\hat{C}_n(X_{n+1}) = \{y \in \mathcal{Y} : R_j^{n+1} \leq \hat{q}_j\},$$

where $x_i^{n+1} = c_j$, and R_j^{n+1} is the conformal score of $X_{n+1} = (x_1^{n+1}, ..., x_s^{n+1})$. This ensure the lower bound defined above (7).

Advanced Group-Balanced Conformal Prediction. In practice, this method tend to over cover because of the drastic size reduction of the calibration set, splitted in m. Some researchers in the article (11; 12) introduces new algorithm solutions, one of them is SAGCCI (Surrogate-Assisted Group-Clustered Conformal Inference) (12). Their first approach is to use the same solution as in the article (11), which is to clusters groups whose conformal score distributions are similar, and then performs conformal inference at the cluster level rather than the group level. This allows bigger calibration sets. Their second idea is to leverage surrogate variables for Y to improve estimation. In many applications, such as medicine, researchers and doctors only have access to a small labeled dataset $\mathcal{D}_1 = \{(X_i, S_i, Y_i)\}$ and a larger unlabeled dataset $\mathcal{D}_0 = \{(X_i, S_i, Y_i = NA)\}$, where the outcome Y is missing but surrogate variables S are observed. These surrogates, which are correlated with Y, allow the method to extract additional information and improve the estimation of the conformal threshold, and the conformal set.

Empirical results show that SAGCCI achieves both tighter prediction sets and low coverage disparity across the groups. However, its guarantees rely on several non-trivial assumptions, including accurate clustering of groups (12).

5.3.2 Class-Conditional Conformal Prediction

In class-conditional Conformal Prediction we try to solve the necessity of having the lower bound coverage guarantee per class of the output y, when $y \in \mathcal{Y} = \{c_1, ..., c_m\}$. Therefore, we want the following guarantee,

$$\mathbb{P}\big(Y_{n+1} \in C(X_{n+1}) \,\big|\, Y_{n+1} = c_j\big) \geq 1 - \alpha$$

for any class c_j for $j = 1, ..., m$. Like in group-balanced CP, we treat each m class in k as a separate calibration set, and apply the conformal algorithm m times, to obtain $\hat{q}_1, ..., \hat{q}_m$ different quantiles. Nonetheless, there is a major difference, which is that we don't know Y_{n+1} true class at inference time, thus we have to slightly change our approach. At inference time, we try each possible $y \in \mathcal{Y}$, and we include y in the set $\hat{C}_n(X_{n+1})$ if $R_j^{n+1} \leq \hat{q}_j$ for all $j = 1, ..., m$

$$\hat{C}_n(X_{n+1}) = \{y \in \mathcal{Y} : R_j^{n+1} \leq \hat{q}_j\}, \quad \text{for any } j = 1, ..., m$$

This ensure the lower bound guarantee (7).

Again, reliable class-conditional CP struggle when only a small number of labeled examples are available per class. To address this limitation, the researcher in (11) introduced clustered conformal prediction, a method that groups together classes with similar conformity score distributions and performs calibration at the level of these clusters rather than individual classes. Empirical results on experiment they conducted demonstrate that this approach consistently improves both class-conditional coverage and the efficiency of prediction sets compared to existing techniques.

5.4 Conformal Prediction with Missing Values

Missing values is one of the major challenges that a statistician faces when trying to apply statistical methods to real-world problems. It has been studied in many areas of statistics. In the article (13), the authors are the first to our knowledge, to establish guaranties in CP with missing values in the covariates. The authors shows that traditional CP methods tend to undercover conditionally to certain patterns in the missing values, and that missing values introduce heteroskedasticity (13). Moreover, they demonstrate that CP on "impute-then-predict" task, meaning we replace missing values using a function $f(X)$ (e.g. the mean...) is marginally valid. This is the case, regardless of the predictive model, the missing value distribution, and the imputation function. This is still a quite weak assumption, because certain patterns of missing values can still lead to undercovergage. To tackle thatt they introduce a new concept: Mask-Conditional-Validity (MCV), that is, valid coverage conditionally on the mask. For any masked combination of the input, we aim to achieve marginal coverage. The first method is called "CP-MDA with exact masking", where the calibration set is built in order to match the exact mask of the test set. The second method is: "CP-MDA with nested masking", where the calibration set has at least the same mask as the test set, but can have more missing value if needed. It can be seen as a relaxed version of the first method, and requires supplementary assumptions to ensure validity (13).

Focusing on Quantile Regression applied to CP, the same CQR algorithm as described in section 3.1. they define two new methods. They respect the constraint of MCV under certain conditions, notably the Missing Completely At Random assumption, defined as: for any $m \in \mathcal{M}$, $\mathbb{P}(M = m \mid X) = \mathbb{P}(M = m)$. The naive solution would be to perform a calibration for each mask, but this would fails as mask on the test samples may have a null frequency of appearing in the calibration set. Instead, the methods tackle the challenge using missing data augmentation, where the calibration set is artificially masked to match the mask of the test set (13).

Figure 13 illustrates how the coverage of predictive intervals varies depending on which subset of the features is missing. In this case, the dataset contains 3 covariates ($2^3 - 1 = 7$ possible masks) with 20% of missing values. The results are averaged over 200 independent runs. The most important feature is X_2, and its combination with X_1 leads to the worst

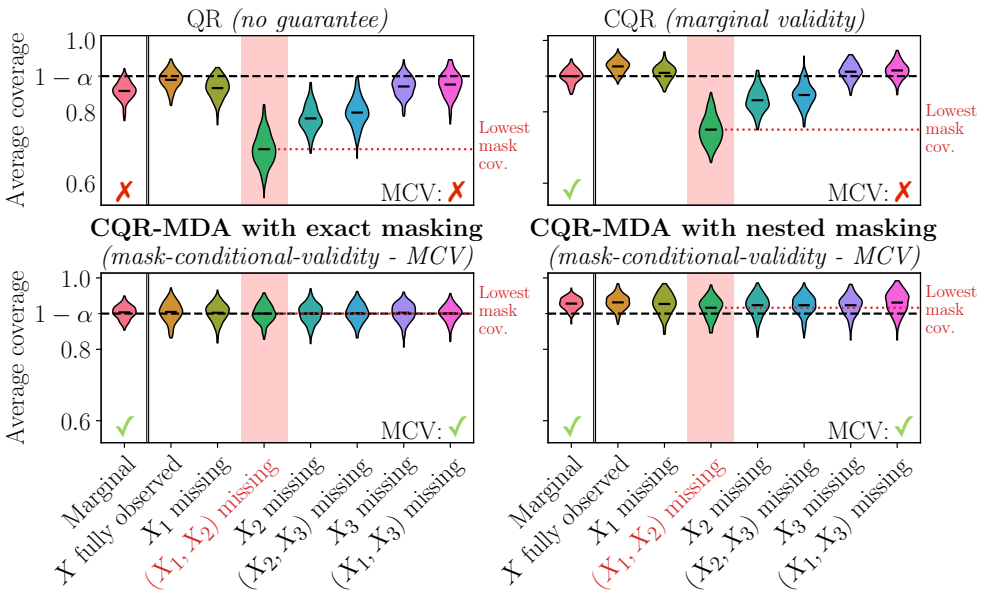

Figure 13: Comparaison of Quantile Regression, CQR, CQR-MDA with exact matching, and CQR-MDA with nested masking over a dataset containing only 3 covariates. (13)

results. This prevent CQR from achieving MCV, while traditional Quantile Regression, as expected, doesn't achieve marginal coverage. Note that MCV is achieved when the worst masked combination of features has an average coverage greater than $1 - \alpha$ (13).

6 Advanced Conformal Prediction Techniques

6.1 Alternatives to SCP and FCP

Cross-conformal prediction (14), CV+, and Jackknife+ (4) provide intermediate solutions between split conformal prediction and full conformal prediction. It balances computational and statistical efficiency (1). Split conformal uses a single data split leading to fast computation but some loss in statistical efficiency. On the other hand, full conformal prediction repeatedly refits the model for every candidate point, achieving high statistical accuracy at a high computational cost. The key idea behind the alternative methods is to reuse the entire dataset for both training and calibration by performing multiple refits on different subsets of the data (e.g., via cross-validation or leave-one-out schemes). This allows them to reduce the variance introduced by data splitting while avoiding the cost of full conformal prediction. To learn more on those methods, you can refer to the articles in the bibliography.

6.2 Detecting Outliers

Detecting outliers is one of the most intuitive applications of Conformal Prediction, as we have a probability threshold over a distribution. Therefore, Conformal prediction can be naturally extended to unsupervised outlier detection. Indeed, given exchangeable calibration data $X_1, \ldots, X_n$ drawn from a distribution, we compute scores $R_i = s(X_i)$ and define a threshold $\hat{q}$ as an empirical quantile of these scores at level $(1 - \alpha)$. For a new test point x, we evaluate its score $s(x)$ and classify it as an outlier if $s(x) > \hat{q}$, and as an inlier otherwise. This procedure guarantees that the probability of falsely declaring a new inlier as an outlier is at most α,

$$\mathbb{P}(C(X_{n+1}) = \text{outlier}) \leq \alpha$$

The effectiveness of the method mainly depends on the choice of the score function, which should assign larger values to points that deviate (1).

This is a simple setting to identify outliers other, more complex approaches have been established. For example, in the article (15), they introduce a new conformal algorithm: Balanced and Conformal Optimized Prediction Sets (BCOPS) that has two main objectives: (1) optimize out-of-sample performance, and (2) detecting outliers. It uses a supervised model $\hat{f}$ to classify the input X_i among k classes. The main interesting fact about this method is that it introduce the possibility that the output can be an empty conformal set $C(X) = \{\}$. An empty set means that the input is classified as an outlier by the model (15). It also introduces some specific metrics for outlier detection.

6.3 Conformal Risk Control

Conformal Risk Control (CRC) was introduced in (16), it extend CP to cases where we are interested in a loss function, instead of miscoverage. Indeed, most of machine learning can be reduced to the minimization (or maximization) of an objective function, often defined as a loss function. The notion of miscoverage is less interesting, because it is a binary notion for

the fact that $Y_i \in C(X_i)$ or not. In many cases, we want to have more informative metrics (F1-score, Recall, Pecision...). In this specific setting, we define the guarantee for the loss function under CRC, as

$$\mathbb{E}\big[\ell(C(X_{n+1}), Y_{n+1})\big] \leq \alpha$$

for any bounded loss function ℓ that shrinks as $C(X_{n+1})$ increase (16). Note that, Conformal Prediction is based on the guarantee,

$$\mathbb{P}\big(Y_{n+1} \notin C(X_{n+1})\big) \leq \alpha.$$

But, with

$$\ell(C(X_{n+1}), Y_{n+1}) = \mathbf{1}\{Y_{n+1} \notin C(X_{n+1})\},$$

we recovere the CRC guarantee. Therefore, Conformal Risk Control is a generalization of Conformal Prediction, as we have seen it so far.

We consider post-processing the predictions of a model f to construct a set-valued function $C_\lambda(\cdot)$, where the parameter λ controls the level of conservativeness, with larger values producing larger (more conservative) prediction sets. The quality of these predictions is measured a bounded loss function $\ell(C_\lambda(x), y) \in (-\infty, B]$, assumed to be non-increasing in λ, meaning getting more conservative shouldn't worsen the loss. The objective is to select $\hat{\lambda}$ based on calibration data $\{(X_i, Y_i)\}_{i=1}^n$ such that the expected risk is controlled. This is reformulated by introducing an exchangeable sequence of non-increasing random functions $L_i : \Lambda \to (-\infty, B]$, where typically $L_i(\lambda) = \ell(C_\lambda(X_i), Y_i)$. The goal is to choose $\hat{\lambda}$ using the first n samples so that $\mathbb{E}[L_{n+1}(\hat{\lambda})] \leq \alpha$. Defining the empirical risk $\hat{R}_n(\lambda) = \frac{1}{n}\sum_{i=1}^n L_i(\lambda)$, the algorithm selects

$$\hat{\lambda} = \inf\left\{\lambda : \hat{R}_n(\lambda) \leq \alpha - \frac{B-\alpha}{n}\right\},$$

where the desired risk level upper bound is $\alpha \in (-\infty, B)$, and with $\hat{\lambda} = \lambda_{\max}$ if the set is empty. This choice guarantees risk control on unseen data, and under i.i.d. and continuity assumptions, also ensures a lower bound,

$$\mathbb{E}\big[L_{n+1}(\hat{\lambda})\big] \geq \alpha - \frac{2B}{n+1}.$$

showing the method is not overly conservative (16).

6.4 Online CP: Adaptive conformal inference

In online prediction, we try to predict data sequentially. Therefore, we try to predict at $t+1$, from a sequence $1, ..., t$, using a predictive model $\hat{f}$. CP is particularly suited for this setting, in (17) proposed an approach to Online CP called "Adaptive Conformal Inference" (ACI). Note that adaptive conformal prediction is also a way to deal with distribution shift, while online CP was introduce long before ACI.

First we define a miscoverage events as,

$$\epsilon_t := \begin{cases} 1, & \text{if } Y_t \notin \hat{C}_t(\alpha_t) \\ 0, & \text{otherwise} \end{cases}$$

The conformal set is

$$\hat{C}_t(\alpha_t) := \{y : R_t \leq \hat{q}_n^t\}.$$

Where $\hat{q}_n^t = R_{(\lceil (1-\alpha_t)\,(n+1) \rceil)}$, and the calibration set is obtain from $\mathcal{D}_2 = \{(X_r, Y_r)\}_{1 \leq r \leq t-1}$. Note that $\hat{q}_n^t$ is defined by α_t, instead of α, where α_t is an updated version of α using a gradient descent setting. As usual, R_i for $i = 1, ..., n$, where n is the size of the calibration set, is the a distance score between $\hat{f}_i$ and y_i (ex: $|\hat{f}_i - y_i|$). We fix step size parameter $\gamma > 0$, the standard online update defined in (17) is

$$\alpha_{t+1} := \alpha_t + \gamma(\alpha - \epsilon_t).$$

An alternative to this update is:

$$\alpha_{t+1} = \alpha_t + \gamma \left(\alpha - \sum_{s=1}^{t} w_s \, \epsilon_t \right),$$

where $\{w_s\}_{1 \leq s \leq t} \subseteq [0, 1]$ is a sequence of increasing weights with $\sum_{s=1}^{t} w_s = 1$. Note, that this setting guaranty the usual marginal coverage (lower bound) under certain assumptions the authors shows that it also has an upper bound. They used this algorithm to predict the price volatility of some stocks, the results are presented in Figure 14.

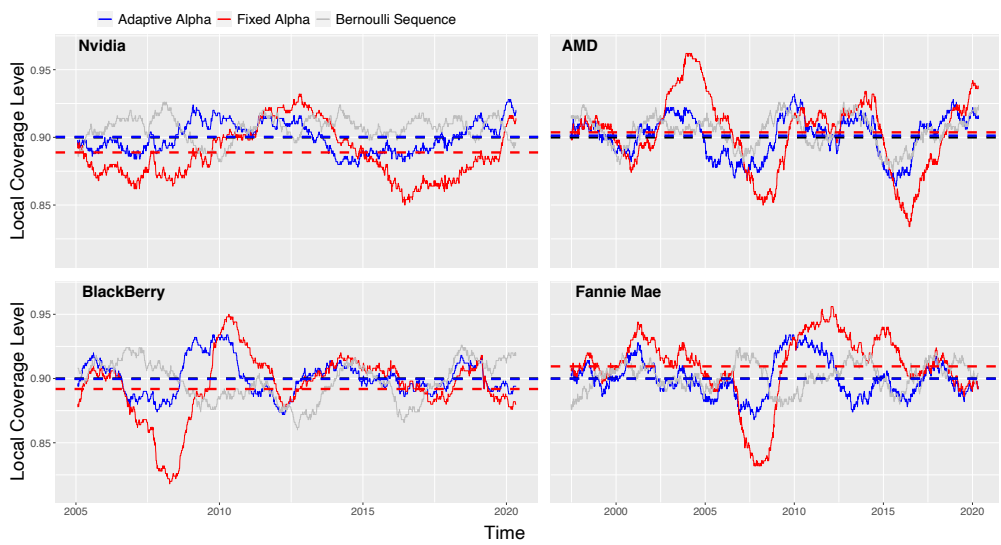

Figure 14: Local coverage rates are shown for adaptive conformal prediction (blue), a non-adaptive approach with fixed α (red), and an i.i.d. Bernoulli(0.1) process (grey) in the context of stock market volatility forecasting. The dotted colored lines represent the average coverage over time for each method, while the black line denotes the target coverage level of $1 - \alpha = 0.9$. The local coverage level is a moving average of the coverage over a centered 500 time step. (17)

Other approaches have been developed, for example in (18), and (19). Those approaches automatically decide for the hyperparameter γ.

6.5 Learning Conformal Classifiers

So far, we have treated Conformal Prediction as an independent procedure from training the model $\hat{f}$. While this have some advantages, like being more flexible, it also has some drawback. Because, any model $\hat{f}$ (π in Figure 15) is trained on being the best at predicting the true output, the set quality is never learned. The article (20) introduce differentiable conformal classification, where the threshold estimate is part of the training, and is adjusted through gradient descent techniques. This is achieved by simulating CP on mini-batches during training, and using smoothing implementation of CP techniques. The conformity score can be smoothed using a sigmoid function. The researcher specifically focuses on Threshold CP (20), and APS (5) for the CP part. More details can be found on the Figure 15.

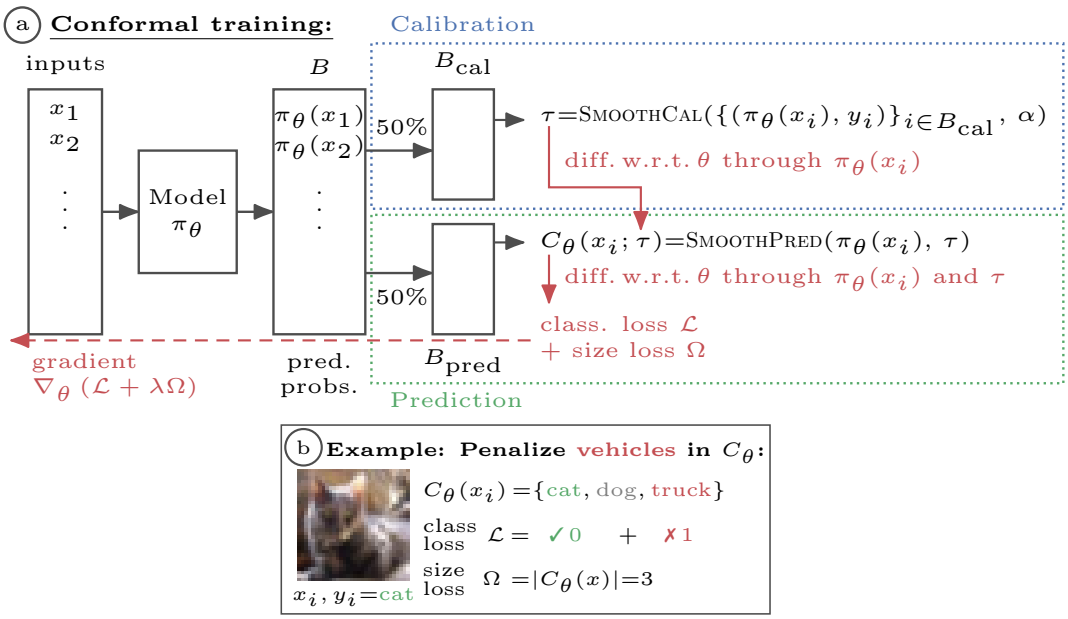

Figure 15: Conformal Training (ConfTr) integrates conformal prediction into training by using differentiable calibration and prediction steps on each mini-batch: one half is used to compute the threshold, the other to form soft confidence sets and optimize a loss on them (20).

The algorithm start by splitting each mini-batch into a calibration set and a prediction set. Then, it uses the calibration set to estimate the threshold τ, ($\hat{q}_n$ in our notation), and build smooth confidence sets on the prediction set. Finally, it compute a loss on these sets and update the model via backpropagation. The researchers (20) shows that this method reduces inefficiency, which is the unnecessary size of the outputted conformal set. After training, it also conserve the coverage guarantee.

6.6 Monte Carlo Conformal Prediction

Monte Carlo Conformal Prediction (MCCP) is a set of new methods using CP on top of the Monte Carlo method. It try to tackle very specific problems. First, the Monte Carlo method

is a set of techniques that aim to approximate a function behavior by first defining the probability distribution of random variables, and to sample from this distribution. Afterward, we use the sampled random variables to compute the function's output.

In this section, we will discuss two main approaches to mix Monte Carlo and Conformal Prediction. One that is focused on tackle label uncertainty when combining multiple sources of labeled data that may differ (21). The other one focus on model robustness in Deep Learning setting using Monte Carlo Dropout (22). In one sentence, the first approach focuses on label randomness, while the second focuses on output randomness. Nonetheless, the principles are the same, and are described in a high view in Figure 16.

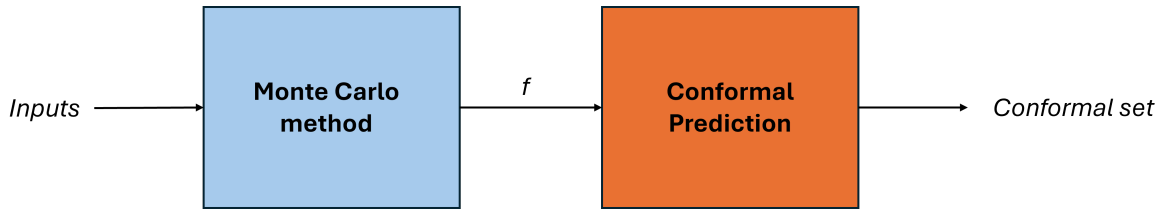

Figure 16: High-level overview of MCCP

Monte Carlo methods applied to CP can lead to a lot of diverse techniques, most of them are probably not yet discovered, here we focus on the two described above.

6.6.1 Monte Carlo Conformal Prediction for ambiguous labels

Standard CP implicitly assumes access to the true label distribution ($P(Y \mid X)$). In practice, labels are often obtained through aggregation of expert annotations (e.g., majority voting), leading to a simplified one-hot distribution (P_{vote}). This simplification ignores inherent ambiguity when experts disagree, which is a common case in many complex task, such as in the medical sector. This result in underestimated uncertainty and empirical under-coverage of true labels (21). This can be observed in a simplified example in Figure 17 (21).
To address this limitation, the paper (21) introduces an alternative approach that models label uncertainty through an aggregated distribution ($P_{\text{agg}}(Y \mid X)$), which captures the diversity of expert opinions instead of collapsing them into a single label. This is achieved by using a specific sampling-based approach. The following is a more technical and detailed explanation.

First, note that this procedure doesn't imply modification on the training data, but focuses only on the calibration set. Therefore, given the calibration set $(X_i, \lambda_i)_{i\in[n]}$, the first step is to sample m labels Y_i^j for each X_i. The sampling is performed according to their respective empirical probability λ_i, where $\mathbb{P}(Y_i^j = k) = \lambda_{ik}$. We obtain m pair for each X_i. The final calibration dataset has size $m \cdot n$, with $(X_i, Y_i^j)_{i\in[n],j\in[m]}$. Finally, we apply the standard split conformal procedure. Because the procedure is defined for the classification task, we obtain the following quantile:

$$\hat{q}_n = R_{(\lfloor \alpha \times m(n+1) \rfloor - m+1)}$$

This method achieves some coverage guaranty that is better described in (21) with respect to the aggregated distribution. Finally, the same article describes another algorithm, called

Empirical CDF MCCP. It is based on the same principles, but with an additional calibration procedure, it comes with other guaranties and performs slightly better on their experiment (21).

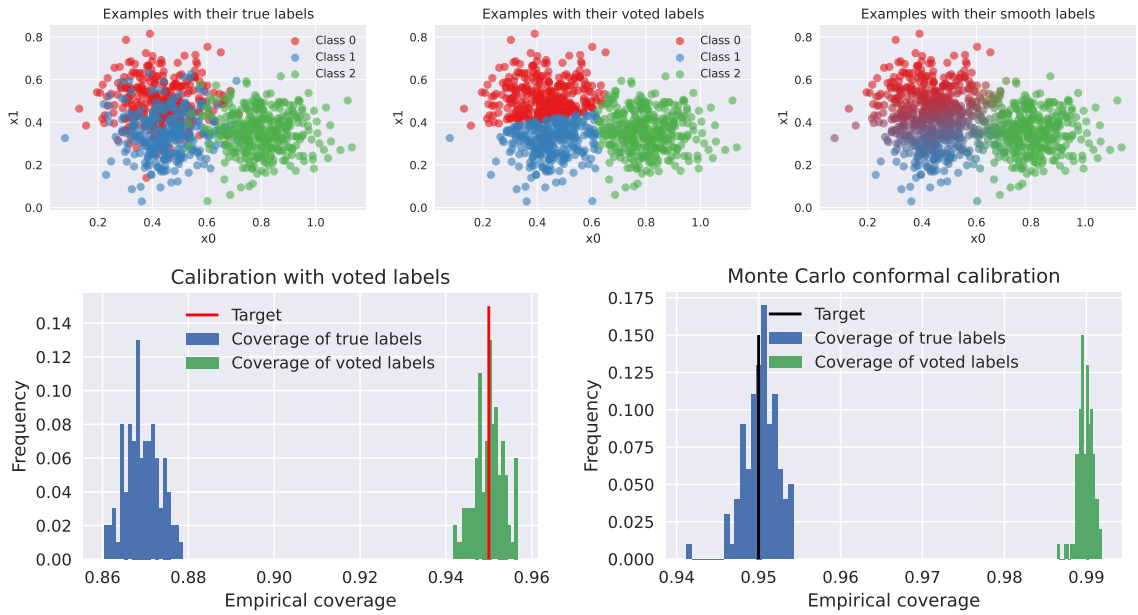

Figure 17: Example of Conformal classification with 3 classes. We observe under-coverage due to label ambiguity. Indeed, with $\alpha = 0.05$, confidence rate should be 95% while it is around 87% in practice using SCP. MCCP close this coverage gap. (21)

6.6.2 Monte Carlo Dropout and Conformal Prediction

Monte Carlo (MC) dropout is a technique used to estimate uncertainty (Uncertainty Quantification [UQ]) in deep learning models by leveraging the standard dropout mechanism at inference time (23). Normally, dropout randomly deactivates neurons during training to prevent overfitting, but it is turned off during testing. In MC dropout, however, dropout remains active during inference, and the model performs multiple forward passes on the same input. Each pass produces a slightly different prediction due to the random dropout. By collecting these multiple predictions, MC dropout builds a distribution of outputs. The final prediction is typically the average (mean) of these outputs, while the variance (or spread) of the predictions quantifies the model's uncertainty (23).

The article (22) introduce adaptive MC dropout, which dynamically determines when enough forward passes have been performed, instead of using a fixed number of passes which is very costly. The method keep track of the variance of predictions and stops early when additional passes provide little new information. This reduces computational cost while maintaining accuracy (22). This new method is coupled with conformal prediction, and can be performed for both deep classification, and deep regression. The process is described in the Figure 18.

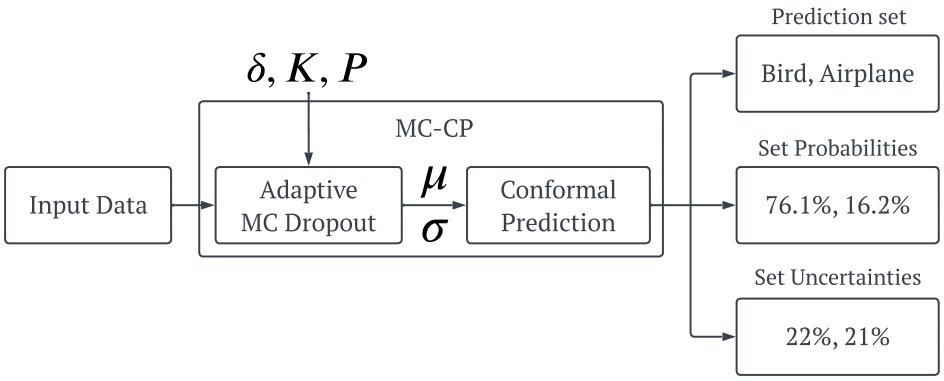

Figure 18: High-level overview of Monte Carlo Dropout and Conformal Prediction (22)

For classification, we first split the data, to obtain a calibration set, and conformal scores are computed to obtain a threshold $\hat{q}_n$, based on the softmax values. Then, in the prediction phase, adaptive MC dropout applied to the predictive model $\hat{f}$ generates the mean and average μ, σ for each classes, for each image. The final output is the conformal set whith (k, μ, σ) per selected classes (22).

The authors show that this new method outperforms standard Conformal methods, such APS and RAPS, as well as, vanilla MC dropout on a wide range of benchmark such as MNIST, CIFAR-10, CIFAR-100...(22).

7 Case studies

7.1 Skin lesion classification

For this use case, we are presenting the results from the article: *Fair Conformal Predictors for Applications in Medical Imaging* (24). They used group-balanced CP combined with APS and RAPS algorithms to a skin lesion classification problem.

The study uses the Fitzpatrick17k dataset, which contains 16,577 dermatological images. The dataset covers 114 skin conditions grouped into three categories: non-neoplastic, benign, and malignant lesions. Each image is also annotated with a Fitzpatrick skin type (a 6-point scale for skin pigmentation), which is used as a grouping variable. The central problem addressed is fairness in medical prediction. Indeed, the prevalence and appearance of conditions like melanoma vary across skin types, the goal is to evaluate whether predictive models maintain consistent performance and reliability across these subgroups, focusing on critical malignant cases (11 categories across the 114, and 13.7% of the dataset). The exact distribution is presented in the Figure 19, with a color tone reflecting the true pigmentation.

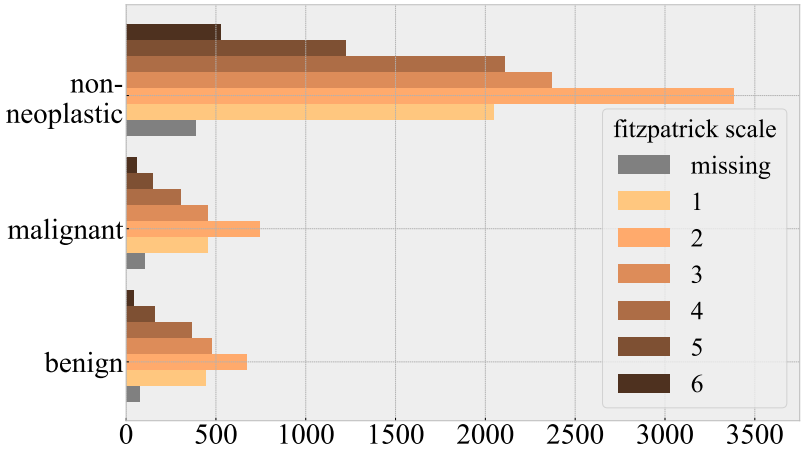

Figure 19: Distribution of the different skin types and classes. (24)

To tackle this, the study compares several conformal prediction methods: Naive (non-conformal baseline), APS, RAPS, GAPS, and GRAPS. The methodological innovation lies in group-aware approaches (GAPS and GRAPS). This allows the model to adapt its uncertainty estimates per subgroup, rather than relying on aggregate calibration that may mask disparities. Performance is evaluated based on coverage (whether the true label is in the prediction set), coverage disparity, and set size disparity. Coverage disparity is the average pairwise difference in marginal coverage,

$$\frac{1}{|k|} \sum_{a,b \in \binom{k}{2}} \left| \text{coverage}_{k=a} - \text{coverage}_{k=b} \right| .$$

where $\binom{k}{2}$ is all pairwise combinations of subgroups.

and the set size disparity is the average pairwise difference in set size:

$$\frac{1}{|k|} \sum_{a,b \in \binom{k}{2}} |\text{set size}_{k=a} - \text{set size}_{k=b}| \,.$$

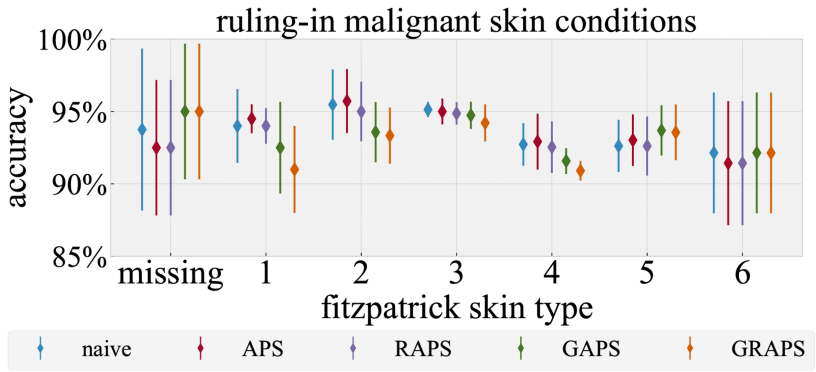

Figure 20: Coverage results for the different subgroup and methods. (24)

The results in Table 4 show that group-aware methods (GAPS and GRAPS) reduce coverage disparity across subgroups compared to standard aggregate methods (APS and RAPS), meaning they are more equitable across skin types. However, this comes at the cost of increased variability in set sizes between groups. Overall, as shown in Figure 20 all methods perform similarly in terms of coverage at a significance level of $\alpha = 0.1$, but group-based methods better capture subgroup-specific uncertainty, highlighting a trade-off between fairness in coverage and consistency in conformal set size.

	method	0.1	0.2	0.3	0.4	0.5
coverage	NAIVE	0.026	0.040	0.059	0.070	0.080
	APS	0.026	0.035	0.048	0.056	0.059
	RAPS	0.026	0.037	0.049	0.054	0.057
	GAPS	0.024	0.029	0.037	0.038	0.045
	GRAPS	0.022	0.028	0.037	0.035	0.039
set size	NAIVE	3.9	2.2	1.2	0.6	0.2
	APS	3.5	3.1	2.4	1.9	1.4
	RAPS	3.6	3.0	2.4	1.8	1.3
	GAPS	9.7	8.1	5.6	4.2	2.8
	GRAPS	9.1	8.0	5.5	4.2	2.7

Table 4: Subgroup disparity in coverage and set size between Fitzpatrick skin types at five different levels of α. (24)

7.2 LLMs for MCQ Answering

Large Language Models (LLMs) have demonstrated strong performance across a wide range of natural language processing tasks, including question answering, summarization, and code

generation. However, a major limitation of these models is their tendency to produce overconfident and sometimes incorrect outputs, commonly referred to as hallucinations. This lack of reliable uncertainty estimation makes them difficult to deploy in high-stakes domains such as healthcare, law, and decision support systems. The paper "*Conformal Prediction with Large Language Models for Multi-Choice Question Answering*" (25) addresses this issue by introducing conformal prediction as a method for providing statistically valid uncertainty estimates for MCQ answering, without requiring retraining or modification of the underlying model (which is very expensive to do for LLMs).

The experiment uses the LLaMA-13B model to perform multiple-choice question answering (MCQA). In this setup, MCQA is formulated as a supervised classification problem, where the goal is to select the correct answer among four options (A, B, C, and D). For each question, the model produces logits corresponding to each answer choice, which are then normalized using the softmax function to obtain probability scores. The dataset used is a subset of MMLU, a benchmark covering 57 domains such as STEM, humanities, and medicine. From this benchmark, 16 subjects are selected, with 100 questions or more each (50% split for the calibration and test set). These subjects are grouped into three categories: computer science, medicine, and business. Predictions are generated using a one-shot approach. The results are presented in figure 21.

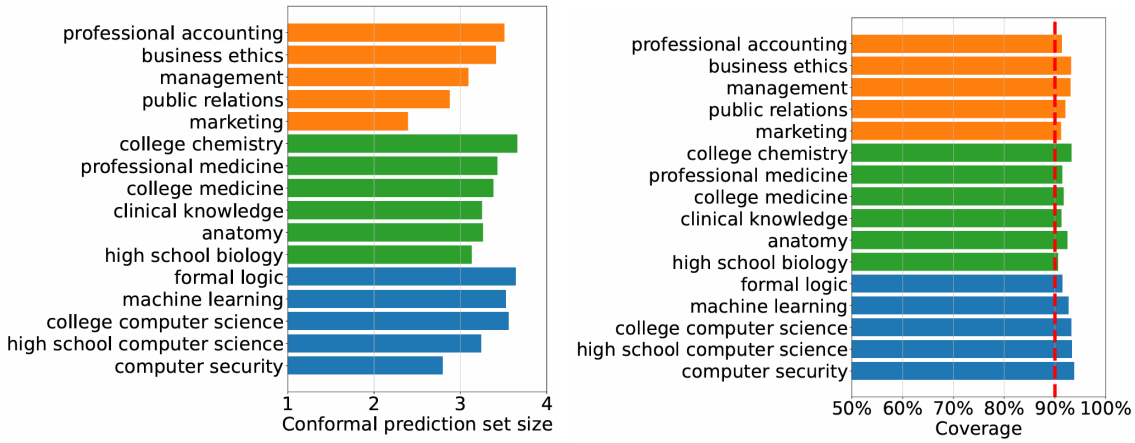

Figure 21: The desired coverage is achieved across all subjects. The red dashed line represents the target coverage level, defined by $1 - \alpha = 0.9$. (25)

The model's probabilities are reasonably well calibrated on average, in this setting we observes that the conformal set size is often around 3 to 4, which means that we regularly output the whole answer space.

To conclude, this article helps on the problematic of trustworthiness in LLMs, leveraging CP techniques.

7.3 Forecasting French electricity spot prices

This case study comes from the article: "*Adaptive Conformal Predictions for Time Series*" (18). It is based on the method from section 6.4. The study focuses on forecasting French

electricity spot prices using an hourly dataset covering 2016–2019, totaling 35,064 observations. Prices are determined daily through an auction market: before 12 a.m., producers and suppliers submit a price at wich they accept to sell for the 24 hours of day D+1, and an algorithm determines the next day's hourly prices (given their bids and other constraints). The prediction task is therefore to forecast, on day D, the full set of 24 prices for day D+1. The model uses explanatory variables presented in table 5. The main challenge comes from the fact that the market can reach high peaks, as observed in figure 23.

Table 5: Extract of the built data set, for French electricity spot price forecasting. (18)

Date and time	Price	Price D-1	Price D-7	For. cons.	DOW
11/01/16 0PM	21.95	15.58	13.78	58800	Monday
11/01/16 1PM	20.04	19.05	13.44	57600	Monday
⋮	⋮	⋮	⋮	⋮	⋮
12/01/16 0PM	21.51	21.95	25.03	61600	Tuesday
12/01/16 1PM	19.81	20.04	24.42	59800	Tuesday
⋮	⋮	⋮	⋮	⋮	⋮
18/01/16 0PM	38.14	37.86	21.95	70400	Monday
18/01/16 1PM	35.66	34.60	20.04	69500	Monday
⋮	⋮	⋮	⋮	⋮	⋮

A separate model is trained for each of the 24 hours using Random Forest to capture hourly specific patterns. The predictions are for 2019, and are obtained with a 3-year sliding window: 1.5 years for training and 1.5 years for calibration. The window moves forward over time.

They use 5 conformal algorithms, Online Sequential Split Conformal Prediction (OSSCP), Offline SSCP, Ensemble Prediction Interval (EnbPI V2), Adaptative Conformal Inferance (ACI) the method described in Section 6.4, and AgACI (A new method introduced in the paper, based on ACI). OSSCP is an online version of split conformal prediction, where the regression model is refit and recalibrated each time with the most recent data. To respect the temporal structure, a sequential split is applied. EnPI builds multiple predictors via bootstrap and updates conformity scores online to adapt intervals over time, without refitting the model. The figure 22 illustrate the Train/Calibration/Test split for each methods. More explanation can be found in the article (18).

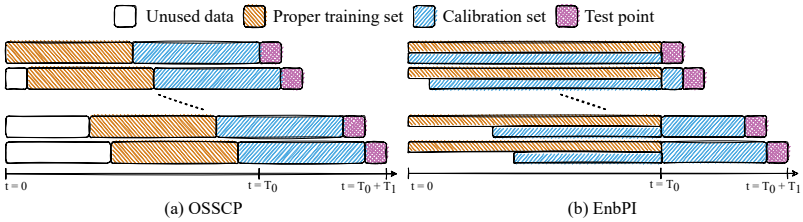

Figure 22: Presenting the OSSCP and EnbPI methods. (18)

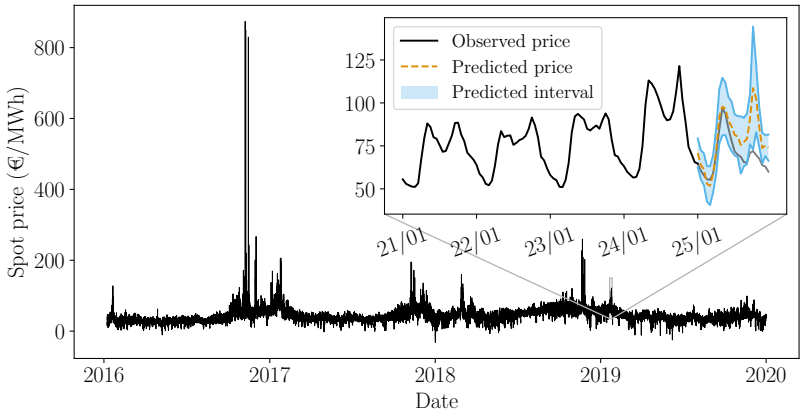

Figure 23: Example of a prediction using AgACI on French electricity spot prices, from 2016 to 2019. (18)

The Figure 24, present the results for each methods, for a coverage rate of 90% ($\alpha = 0.1$). All the methods offer the required empirical coverage, except ACI for a specific γ. This illustrates the sensitivity of this method to the choice of its hyperparameter. The tighter coverage is also achieved by the ACI method with the right γ, followed by AgACI.

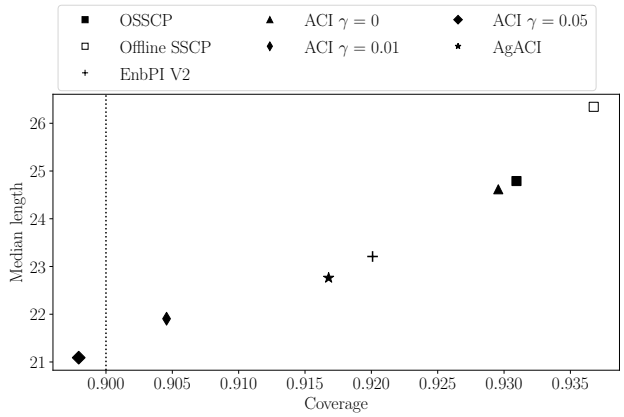

Figure 24: The performance of each methods on the spot price prediction task. (18)

8 Conclusion

Conformal Prediction address one of the central challenges of modern machine learning: quantifying uncertainty with reliable guarantees. In contrast to traditional approaches that rely heavily on distributional assumptions or asymptotic arguments, conformal methods offer finite-sample, distribution-free guarantees under the mild assumption of exchangeability. Throughout this report, we have explored the theoretical foundations of conformal prediction, beginning with the quantile-based intuition. We then examined practical implementations, notably Split Conformal Prediction (SCP) and Full Conformal Prediction (FCP). SCP emerges as a practical and widely used approach, while FCP provides a conceptually interesting, but computationally intensive alternative. We also discussed several important extensions that enhance the flexibility and applicability of conformal methods. Techniques such as conformalized quantile regression, weighted conformal prediction, and classification-based approaches. Furthermore, advanced topics includes Mondrian conformal prediction, covariate shift adaptation, distribution drift, and handling missing data. Those techniques demonstrate how the framework can be extended beyond idealized assumptions to more realistic scenarios. Empirical applications across medical diagnosis, energy forecasting, and large language models confirm that conformal prediction translates theoretical guarantees into real-world applications. As the field continue to grow, it is well-positioned to become a foundational component of trustworthy AI systems.

8.1 Future perspectives

Conformal Prediction has a promising future in a wide range of research areas. Here I present some I believe may have a particularly interesting perspective. I categorized them in two groups:

- **Applications**: Conformal Prediction has become a popular tool when one needs to produce prediction sets with coverage guarantees. In recent years, it has been applied across a wide range of fields and industries. Nonetheless, there is still significant room for new applications and for solutions tailored to specific problems. Among others, these fields include medical sciences, finance, robotics, social sciences, and LLMs. Medical sciences have long been a cornerstone of applied statistics, and conformal prediction has much to offer given the strong need for reliable decision. Financial engineering is also a promising area due to its reliance on predictive models and its need to hedge against or capitalize on extreme events. In particular, the field of derivative products appears promising, it would allow to design strategies based on coverage guarantees. Robotics-based decision systems require guarantees that models behave reliably under distributional shifts, making conformal prediction highly relevant. In social sciences, it can enable more granular assessments while helping to mitigate certain biases. Finally, conformal prediction could be used to improve the reliability of LLM by providing clearer guarantees.

- **Theoretical bridges**: Conformal prediction can be applied to, or further developed within, other research areas that could benefit from its guarantees. These include

reinforcement learning, optimal transport, or world models. Reinforcement learning is based on selecting actions that maximize a reward function, but in practice, the optimal path can be uncertain or ambiguous. A potential application of conformal prediction would be to introduce stochasticity by selecting actions from a conformal set with a predefined coverage guarantee. Optimal transport is another promising area where connections with conformal prediction could be explored, particularly in modeling uncertainty over distributions. Finally, world models could also benefit from conformal prediction. For instance, by producing sets of predictions that remain physically consistent.

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A Additional Material

A.1 SCP Mean Regression algorithm

Algorithm 1 Mean Conformal Regression

Require: Training set $\mathcal{D}_1 = \{(X_i, Y_i)\}_{i=1}^{n_1}$, calibration set $\mathcal{D}_2 = \{(X_i, Y_i)\}_{i=1}^{n_2}$, miscoverage level $\alpha \in (0, 1)$

Ensure: Prediction interval $\hat{C}_{n_2}(x)$ for a new point x

1: Fit the regression model $\hat{f}_{n_1}$ on $\mathcal{D}_1$

2: **for** $i = 1, \ldots, n_2$ **do**

3: Compute calibration residual

$$R_i = |Y_i - \hat{f}_{n_1}(X_i)|$$

4: **end for**

5: Sort the scores:

$$R_{(1)} \leq R_{(2)} \leq \cdots \leq R_{(n_2)}$$

6: Define

$$\hat{q}_{n_2} = R_{(\lceil (1-\alpha)(n_2+1) \rceil)}$$

with the convention $R_{(n_2+1)} = \infty$

7: For a new input x, output

$$\hat{C}_{n_2}(x) = \left[\hat{f}_{n_1}(x) - \hat{q}_{n_2},\ \hat{f}_{n_1}(x) + \hat{q}_{n_2}\right]$$

A.2 SCP CQR algorithm

Algorithm 2 Conformalized Quantile Regression (CQR)

Require: Training set $\mathcal{D}_1 = \{(X_i, Y_i)\}_{i=1}^{n_1}$, calibration set $\mathcal{D}_2 = \{(X_i, Y_i)\}_{i=1}^{n_2}$, miscoverage level $\alpha \in (0, 1)$

Ensure: Prediction interval $\hat{C}_{n_2}(x)$ for a new point x

1: Fit the lower quantile regression model $\hat{f}_{n_1}^{\alpha/2}$ on $\mathcal{D}_1$

2: Fit the upper quantile regression model $\hat{f}_{n_1}^{1-\alpha/2}$ on $\mathcal{D}_1$

3: **for** $i = 1, \ldots, n_2$ **do**

4: Compute the calibration score

$$R_i = \max\left\{\hat{f}_{n_1}^{\alpha/2}(X_i) - Y_i,\ Y_i - \hat{f}_{n_1}^{1-\alpha/2}(X_i)\right\}$$

5: **end for**

6: Sort the scores:

$$R_{(1)} \leq R_{(2)} \leq \cdots \leq R_{(n_2)}$$

7: Define the conformal correction term

$$\hat{q}_{n_2} = R_{(\lceil (1-\alpha)(n_2+1) \rceil)}$$

with the convention $R_{(n_2+1)} = \infty$

8: For a new input x, output the conformalized quantile prediction interval

$$\hat{C}_{n_2}(x) = \left[\hat{f}_{n_1}^{\alpha/2}(x) - \hat{q}_{n_2},\ \hat{f}_{n_1}^{1-\alpha/2}(x) + \hat{q}_{n_2}\right]$$

A.3 Weighted SCP algorithm

Algorithm 3 Locally Weighted Conformal Regression

Require: Training set $\mathcal{D}_1 = \{(X_i, Y_i)\}_{i=1}^{n_1}$, calibration set $\mathcal{D}_2 = \{(X_i, Y_i)\}_{i=1}^{n_2}$, miscoverage level $\alpha \in (0, 1)$

Ensure: Prediction interval $\hat{C}_{n_2}(x)$ for a new point x

1: Fit a regression model $\hat{f}_{n_1}$ on $\mathcal{D}_1$

2: Fit a model $\hat{\rho}_{n_1}$ on $\mathcal{D}_1$ to estimate the conditional mean absolute deviation (MAD) of $Y - \hat{f}_{n_1}(X)$ given X

3: **for** $i = 1, \ldots, n_2$ **do**

4: Compute the weighted conformity score

$$R_i = \frac{|Y_i - \hat{f}_{n_1}(X_i)|}{\hat{\rho}_{n_1}(X_i)}$$

5: **end for**

6: Sort the scores:

$$R_{(1)} \leq R_{(2)} \leq \cdots \leq R_{(n_2)}$$

7: Define the conformal correction term

$$\hat{q}_{n_2} = R_{(\lceil (1-\alpha)(n_2+1) \rceil)}$$

with the convention $R_{(n_2+1)} = \infty$

8: For a new input x, output the locally weighted conformal prediction interval

$$\hat{C}_{n_2}(x) = \left[\hat{f}_{n_1}(x) - \hat{\rho}_{n_1}(x)\hat{q}_{n_2},\ \hat{f}_{n_1}(x) + \hat{\rho}_{n_1}(x)\hat{q}_{n_2}\right]$$

A.4 Vanilla Split Conformal Classification algorithm

Algorithm 4 Vanilla Split Conformal Classification

Require: Training set $\mathcal{D}_1 = \{(X_i, Y_i)\}_{i=1}^{n_1}$, calibration set $\mathcal{D}_2 = \{(X_i, Y_i)\}_{i=1}^{n_2}$, miscoverage level $\alpha \in (0, 1)$

Ensure: Prediction set $\hat{C}_{n_2}(x)$ for a new point x

1: Fit a classifier $\hat{f}_{n_1}$ on $\mathcal{D}_1$

2: $\hat{f}_{n_1}(x; k)$ denotes the score (e.g., probability) of class k

3: **for** $i = 1, \ldots, n_2$ **do**

4: Compute conformity score using the true label:

$$R_i = \hat{f}_{n_1}(X_i; Y_i)$$

5: **end for**

6: Sort the scores:

$$R_{(1)} \leq R_{(2)} \leq \cdots \leq R_{(n_2)}$$

7: Define the threshold

$$\hat{q}_{n_2} = R_{(\lfloor \alpha(n_2+1) \rfloor)}$$

with the convention $R_{(n_2+1)} = 1$

8: For a new input x, output the prediction set

$$\hat{C}_{n_2}(x) = \{\, k \in \mathcal{Y} : \hat{f}_{n_1}(x; k) \geq \hat{q}_{n_2} \,\}$$

A.5 APS algorithm

Algorithm 5 Adaptive Prediction Sets (APS)

Require: Training set $\mathcal{D}_1 = \{(X_i, Y_i)\}_{i=1}^{n_1}$, calibration set $\mathcal{D}_2 = \{(X_i, Y_i)\}_{i=1}^{n_2}$, miscoverage level $\alpha \in (0, 1)$

Ensure: Prediction set $\hat{C}_{n_2}(x)$ for a new point x

Fit a classifier $\hat{f}_{n_1}$ on $\mathcal{D}_1$

For each input x, let $\hat{f}_{n_1}(x; k)$ denote the score of class k

for $i = 1, \ldots, n_2$ **do**

Sort the class scores in decreasing order:

$$\hat{f}_{n_1}(X_i; k_{(1)}) \geq \hat{f}_{n_1}(X_i; k_{(2)}) \geq \cdots \geq \hat{f}_{n_1}(X_i; k_{(c)})$$

Let t_i be the rank such that $k_{(t_i)} = Y_i$

Compute the conformity score

$$R_i = \sum_{j=1}^{t_i} \hat{f}_{n_1}(X_i; k_{(j)})$$

end for

Sort the conformity scores:

$$R_{(1)} \leq R_{(2)} \leq \cdots \leq R_{(n_2)}$$

Define the conformal threshold

$$\hat{q}_{n_2} = R_{(\lceil (1-\alpha)(n_2+1) \rceil)}$$

For a new input x, sort the class scores in decreasing order:

$$\hat{f}_{n_1}(x; k_{(1)}) \geq \hat{f}_{n_1}(x; k_{(2)}) \geq \cdots \geq \hat{f}_{n_1}(x; k_{(c)})$$

Define

$$\hat{t}(x) = \min \left\{ t \in \{1, \ldots, c\} : \sum_{j=1}^{t} \hat{f}_{n_1}(x; k_{(j)}) \geq \hat{q}_{n_2} \right\}$$

Output the prediction set

$$\hat{C}_{n_2}(x) = \{k_{(1)}, \ldots, k_{(\hat{t}(x))}\}$$

A.6 Full conformal Prediction algorithm

Algorithm 6 Full Conformal Prediction (FCP) for Regression

Require: Training set $\{(X_i, Y_i)\}_{i=1}^n$, test input $X_{n+1} = x$, candidate response space $\mathcal{Y}$, miscoverage level $\alpha \in (0, 1)$

Ensure: Prediction set $\hat{C}_n(x)$

Initialize $\hat{C}_n(x) \leftarrow \varnothing$

for each candidate value $y \in \mathcal{Y}$ **do**

Form the augmented dataset

$$D^{(y)} = \{(X_1, Y_1), \ldots, (X_n, Y_n), (X_{n+1}, y)\}$$

Fit the predictor $\hat{f}_{n+1}^{(y)}$ on $D^{(y)}$

for $i = 1, \ldots, n$ **do**

Compute the conformity score

$$R_i^{(y)} = \left| Y_i - \hat{f}_{n+1}^{(y)}(X_i) \right|$$

end for

Compute the test conformity score

$$R_{n+1}^{(y)} = \left| y - \hat{f}_{n+1}^{(y)}(X_{n+1}) \right|$$

Sort the $n + 1$ conformity scores:

$$R_{(1)}^{(y)} \leq R_{(2)}^{(y)} \leq \cdots \leq R_{(n+1)}^{(y)}$$

Define the conformal threshold

$$\hat{q}_n^{(y)} = R_{(\lceil (1-\alpha)(n+1) \rceil)}^{(y)}$$

if $R_{n+1}^{(y)} \leq \hat{q}_n^{(y)}$ **then**

Update the prediction set:

$$\hat{C}_n(x) \leftarrow \hat{C}_n(x) \cup \{y\}$$

end if

end for

return $\hat{C}_n(x)$